Optimization-Driven Feature Learning for Long-Tailed Object Detection in Optical Remote Sensing Imagery

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ABSTRACT

With the rapid advancement of remote sensing technologies, deep learning–based object detection has been widely adopted for large-scale optical remote sensing imagery. Despite the remarkable progress achieved by convolutional and transformer-based detectors, their performance remains significantly limited by the long-tailed distribution that is inherently present in real-world remote sensing datasets. Existing approaches address this issue primarily from data rebalancing or class-frequency perspectives, while largely overlooking the underlying optimization difficulty that governs the learning behaviour of tail and hard samples. To address this limitation, we propose a novel optimization-driven long-tailed object detection framework for optical remote sensing images, which explicitly regulates training dynamics rather than relying solely on statistical class distributions. The proposed method introduces an Optimization-Driven Feature Emphasis (ODFE) mechanism to adaptively highlight discriminative multi-scale features that are critical for hard and tail samples during optimization. In addition, an Optimization-Balanced Mosaic (OBM) augmentation strategy is designed to increase the effective exposure of optimization-tail samples, alleviating biased gradient updates caused by conventional class-agnostic augmentation. Furthermore, an Optimization-Aware Classification Loss (OACL) is formulated to reweight samples according to their optimization difficulty, ensuring stable convergence and balanced learning across head and tail categories. Extensive experiments conducted on three public optical remote sensing benchmarks, namely DIOR, DOTA, and FAIR1M, demonstrate that the proposed framework achieves competitive or superior performance compared with reproduced baselines evaluated under identical training settings; results from original papers are clearly annotated and interpreted with appropriate caution. In particular, notable performance gains are observed on small objects, background-cluttered categories, and fine-grained long-tailed classes. These results suggest that explicitly modeling optimization difficulty provides a more effective learning strategy for long-tailed object detection in optical remote sensing imagery, within the scope of the experimental settings considered.

Keywords: data augmentation; remote sensing imagery; optimization difficulty; object detection; long-tailed distribution .

Introduction

Remote sensing technology provides large-scale Earth observation through satellite and aerial platforms, and has become indispensable for a wide range of civilian and defence applications, including urban planning, traffic monitoring, environmental assessment, and disaster response[1–4]. With the rapid advancement of high-resolution optical sensors, remote sensing imagery now captures increasingly fine-grained scene details, making accurate and efficient object detection a critical enabling technology for downstream analysis[5,6].

The success of deep learning has driven remarkable progress in object detection on natural images. One-stage detectors such as YOLO[7], SSD[8], and region-based frameworks such as Faster R-CNN[9] have established strong baselines. Subsequent advances incorporating multi-scale feature pyramids, attention mechanisms, and transformer architectures have further improved detection accuracy and robustness[10–12]. These methods have been successfully adapted for remote sensing, achieving notable results on benchmarks such as DIOR[13], DOTA[14], and FAIR1M[15].

The core challenge: long-tailed distributions in remote sensing. Despite these advances, remote sensing object detection faces a fundamental and persistent challenge: *long-tailed class distributions*. In large-scale benchmarks such as DIOR, DOTA, and FAIR1M, a small number of dominant (head) categories account for the vast majority of training instances, while many tail categories contain only sparse annotations[13,14]. This distributional skew leads detectors to be strongly biased towards head categories, resulting in severely degraded performance on tail categories that are often practically the most important targets,

such as helicopters, windmills, and storage tanks.

Existing methods to address long-tailed learning follow one of four paradigms: (1) *data-level rebalancing*[16,17]; (2) *loss reweighting*, such as Focal Loss[10] and Equalization Loss[18]; (3) *classifier decoupling*[19]; and (4) *feature augmentation*[17]. Although effective in constrained settings, these methods share a critical and common assumption: *tail categories are difficult solely because of their low sample frequency*. Recent studies, however, have demonstrated that frequency imbalance alone cannot fully explain poor tail-category performance[20,21], and that a deeper underlying cause must be addressed.

The overlooked cause: optimization difficulty. We argue that the true bottleneck in long-tailed remote sensing detection is not class frequency, but *optimization difficulty*. Tail objects in remote sensing imagery are typically small, densely distributed, and embedded in complex backgrounds, which causes their training gradients to be unstable, their loss surfaces to be highly non-convex, and their convergence to be slow and erratic[22,23]. Critically, this optimization challenge is *not* correlated with class frequency: many tail categories can be relatively easy to optimize under sufficient exposure, while some frequent categories with ambiguous or cluttered appearances remain persistently hard to learn[20,24]. This observation reveals a fundamental mismatch in existing methods, which regulate *what is sampled* based on class frequency rather than *how samples are optimized* in terms of gradient stability and convergence behaviour.

Our approach: a unified optimization-driven framework. Motivated by the above analysis, we propose a unified *optimization-driven* framework whose three modules all share a single per-sample difficulty score $\tau_i^{(t)}$, computed from gradient sensitivity, convergence instability, and local curvature. Figure 2 illustrates how this shared signal simultaneously drives ODFE, OBM, and OACL. Below we state the motivation for each module concisely; full formulations appear in Section .

(1) ODFE — feature emphasis guided by optimization, not appearance. Standard FPN reweighting is appearance-driven and agnostic to which samples are hard to learn. ODFE reweights feature pyramid levels by $\tau_i^{(t)}$, ensuring the most optimization-critical levels receive greater representational emphasis.

(2) OBM — augmentation sampling guided by optimization, not frequency. Class-frequency-based mosaic treats all samples of a class equally. OBM sets per-sample mosaic probability $P_i \propto \tau_i^{(t)}$, increasing training exposure of hard instances regardless of class frequency.

(3) OACL — loss reweighting via multi-dimensional difficulty, not confidence. Focal Loss uses one scalar (confidence); OACL uses the full $\tau_i^{(t)}$ to reweight per-sample classification loss, suppressing easy-sample gradient dominance and amplifying tail-sample contributions.

The three modules are jointly optimised under $\mathscr{L}_{\text{final}} = \mathscr{L}_{\text{loc}} + \lambda \mathscr{L}_{\text{cls}}$, addressing the long-tailed challenge at the feature, data, and loss levels simultaneously. Experiments on DIOR, DOTA, and FAIR1M demonstrate consistent state-of-the-art improvements, with extensive ablation studies validating each component's contribution.

In summary, the main contributions of this work are as follows:

- We reframe long-tailed object detection in remote sensing as an *optimization difficulty* problem rather than a class-frequency problem, offering a complementary perspective on tail-category performance degradation that is distinct from existing data-rebalancing and loss-reweighting approaches.
- We propose *Optimization-Driven Feature Emphasis* (ODFE), which adaptively reweights multi-scale features based on gradient sensitivity and learning instability, ensuring that optimization-critical feature levels are explicitly emphasized for hard and tail samples.
- We introduce *Optimization-Balanced Mosaic* (OBM) augmentation, which dynamically adjusts sample exposure in mosaic composition according to optimization difficulty, effectively increasing the training presence of hard samples regardless of class frequency.
- We formulate *Optimization-Aware Classification Loss* (OACL), which reweights per-sample loss contributions using a multi-dimensional difficulty signal, achieving stable and balanced convergence across head and tail categories.
- We conduct extensive experiments on DIOR, DOTA, and FAIR1M, achieving competitive performance with notable improvements on challenging tail categories and small objects, particularly in controlled comparisons where baselines are reproduced under identical training settings.

Related Work

In this section, we review representative research progress related to our work from three perspectives: (1) long-tailed learning and object detection, (2) object detection in optical remote sensing images, and (3) data augmentation and optimization strategies for long-tailed detection.

Table 1. Structured novelty comparison with related hard-sample mining, adaptive reweighting, and optimization-aware methods. Checkmarks (✓) denote full support; dashes (–) denote absence; ∼ denotes partial support. The proposed method is the only framework that jointly addresses all five dimensions.

Method	Year	Difficulty Signal	Multi-dim. Signal	EMA Noise Robust	Feature Regulation	Augment. Control	Loss Reweight	Unified Pipeline	RS-Specific
Focal Loss[10]	2017	Classification confidence	–	–	–	–	✓	–	–
OHEM[31]	2016	Loss ranking	–	–	–	–	✓	–	–
GHM[32]	2019	Gradient magnitude hist.	–	∼	–	–	✓	–	–
EQL[18]	2020	Class frequency	–	–	–	–	✓	–	–
BAGS[33]	2020	Class frequency (grouped)	–	–	–	∼	✓	–	–
SimDE[20]	2023	Semantic difficulty	∼	–	–	–	✓	–	–
CubAL[34]	2023	Curriculum loss	–	–	–	✓	✓	–	–
ABNet[35]	2022	Attention statistics	–	–	✓	–	–	–	✓
Ours (ODFE+OBM+OACL)	2024	**Grad+Fluct+Curve**	✓	✓	✓	✓	✓	✓	✓

Long-Tailed Learning and Object Detection

Long-tailed learning has attracted increasing attention due to the inherent imbalance present in real-world data distributions, where a few classes predominate in the training samples whereas the rest of classes are severely under-represented[25]. Early studies systematically analyzed the challenges caused by data imbalance and demonstrated that conventional empirical risk minimization tends to bias decision boundaries toward head classes[26].

Recent works have explored long-tailed recognition from various perspectives. Decoupling-based methods separate representation learning and classifier optimization to alleviate classifier bias under long-tailed distributions[19]. Class-balanced loss functions and reweighting strategies attempt to compensate for imbalance by adjusting loss contributions according to effective sample numbers[17]. Focal loss and its variants further emphasize hard samples by down-weighting easy negatives[10,27].

On top of classification, the additional challenges in long-tailed object detection include spatial localization, background clutter, and scale variation. Several approaches extend long-tailed learning to detection via adaptive classification heads or reweighted sampling[18,28]. However, these methods rely primarily on class-frequency statistics and implicitly assume that rare samples are always hard to optimise[29].

More recent research demonstrates that rarity and optimization difficulty are not equivalent: some frequent categories converge slowly due to visual complexity, while some tail classes can be relatively straightforward to optimise[20,30]. This fact inspires studies that put emphasis on optimization behaviour as opposed to only data distribution. However, in practice, the vast majority of current long-tailed detection methods do not formalise the optimization difficulty in training, and there is still a lot of room to optimize complex remote sensing tasks.

Although conceptually related to prior hard-sample mining and adaptive reweighting methods, our approach is *fundamentally different* in its formulation, scope, and target domain. Table 1 provides a structured comparison with the most closely related methods.

Key distinctions from existing methods. Three fundamental properties differentiate our framework from prior work:

(1) Multi-dimensional vs. single-dimensional difficulty. All existing hard-sample mining methods — Focal Loss, OHEM, GHM, EQL — rely on a *single scalar* at a *single time instant* to characterise sample difficulty: classification confidence, loss magnitude, or gradient histogram. Such scalar signals are inherently limited because they conflate multiple sources of difficulty (appearance ambiguity, background clutter, scale variation, annotation noise) into one number, which cannot distinguish samples that are genuinely hard to optimise from samples that are momentarily uncertain. Our optimization difficulty metric $\tau_i^{(t)}$ is the first to jointly capture *three orthogonal dimensions*: gradient sensitivity $g_i^{(t)}$ (how strongly the sample perturbs the loss landscape), convergence instability $\delta_i^{(t)}$ (whether the loss is reducing smoothly), and curvature proxy $\kappa_i^{(t)}$ (how rapidly the gradient direction changes). This richer signal enables substantially more reliable identification of genuinely hard tail samples.

(2) Single-level vs. unified cross-level integration. Prior methods apply their difficulty signal at exactly one level: loss reweighting (Focal Loss, OHEM, GHM, EQL), augmentation sampling (CubAL, BAGS), or feature attention (ABNet). No prior work propagates a single unified difficulty signal simultaneously across all three levels. Our framework is unique in using the *same* $\tau_i^{(t)}$ to simultaneously drive feature emphasis (ODFE), augmentation sampling (OBM), and loss reweighting (OACL). This creates a mutually reinforcing pipeline: ODFE improves the quality of the difficulty estimate by learning better representations; OBM increases training exposure for hard samples; OACL ensures their gradients dominate the classification objective. The three modules cannot be applied independently at full effectiveness.

(3) General domain vs. remote-sensing-specific design. Existing methods are designed for natural image datasets (ImageNet, COCO, LVIS) where difficulty correlates with low classification confidence. In remote sensing, tail objects are difficult for structurally different reasons: small physical size relative to scene, cluttered backgrounds with high spectral similarity to non-targets, dense spatial layouts causing annotation overlap, and arbitrary orientation. These properties produce

optimization instability (persistent gradient variance and loss fluctuation) rather than low confidence. Our difficulty metric is explicitly designed to capture this instability, making it directly applicable to the remote sensing setting without heuristic tuning.

Object Detection in Optical Remote Sensing Images

The problem of object detection in optical remote sensing images has been widely researched because of its significance in the areas of applications like urban planning, traffic monitoring and in military reconnaissance[36]. The remote sensing images present special difficulties as compared to natural images, such as large-scale variation, saturated object density, intricate backgrounds, and extreme aspect ratios [6,12].

Initial techniques were based on hand-crafted characteristics and sliding-window techniques [37,38]. The success of deep learning encouraged the use of CNN-based detectors in remote sensing imagery, including Faster R-CNN and SSD, which were adapted accordingly to these images [39,40]. Later papers proposed feature pyramid networks and context modelling to enhance multi-scale representation[41,42].

YOLO has become popular because one-stage detectors are efficient, especially the YOLO family. A number of YOLO variants, specific to remote sensing, have been suggested to improve background classification and small objects detection better [43,44]. Transformer based architectures and attention mechanisms have also been used to enhance global context modelling and long-range dependency capture further [45,46].

Although these improvements have been made, the majority of remote sensing detectors are trained using conventional optimization pipelines. These models tend to have unstable training dynamics that cause poor tail category generalisation when dealing with long-tailed distributions, and have been applied to address various problems such as reinforcement learning and computer-assisted instruction [47]. This shortcoming underscores the need to incorporate long-tailed learning concepts in remote sensing detection models.

ABNet[35] proposes an Adaptive Balanced Network for multiscale object detection in remote sensing imagery. The key contribution of ABNet is a feature-level balancing mechanism that adaptively aggregates multi-scale features by learning channel-wise and spatial attention weights, enabling the detector to better represent objects across a wide range of scales and categories. ABNet demonstrates that representational imbalance across feature pyramid levels is a significant obstacle to detecting both large and small remote sensing objects and that explicitly balancing feature contributions improves detection of underrepresented categories. While ABNet addresses the *representational* side of the imbalance problem through adaptive feature aggregation, it does not model the *optimization dynamics* of individual training samples. Our ODFE module is complementary to ABNet: rather than reweighting features based on attention statistics, ODFE reweights feature levels based on gradient sensitivity and learning instability, explicitly targeting samples that are hard to optimize rather than those that are merely underrepresented at the feature level.

Dual-Perspective Alignment[48] proposes a dual-perspective alignment learning framework for multimodal remote sensing object detection. The method aligns visual and semantic feature spaces from two complementary modalities through a cross-modal alignment module, reducing the representation gap between optical and non-optical sensing data and improving detection performance on fine-grained and rare object categories. Dual-Perspective Alignment highlights that leveraging complementary information sources and aligning heterogeneous feature distributions can substantially improve tail-category recognition in remote sensing. Our work shares the motivation of improving tail-category detection but approaches the problem from a fundamentally different angle: rather than aligning multi-modal feature spaces, we model the optimization difficulty of each training sample and use this signal to simultaneously regulate feature emphasis, augmentation, and loss reweighting within a single-modality framework.

PointSAM[49] proposes a pointly-supervised adaptation of the Segment Anything Model (SAM) for remote sensing detection and segmentation, achieving competitive performance with only point-level annotations. This work highlights a critical challenge directly relevant to our setting: in long-tailed remote sensing datasets, tail categories suffer from both label scarcity and annotation sparsity — the same categories that are rare are often annotated with lower precision. PointSAM's label-efficient approach is complementary to ours: while PointSAM addresses annotation efficiency, our ODFE+OBM+OACL pipeline addresses the optimization difficulty that persists even when sufficient annotations exist. Together, these directions underscore the need for methods robust to both data scarcity and distributional imbalance in RS detection.

Semi-supervised Object Detection with Uncurated Unlabeled Data[50] exploits large-scale unlabeled RS imagery to improve detection performance under limited supervision. The method demonstrates that data-centric strategies — leveraging unlabeled data to augment the effective training distribution — can substantially improve detection of rare categories. This work is orthogonal and complementary to our optimization-centric approach: where semi-supervised methods expand the effective data distribution, our framework improves *how* the available labeled data is exploited during optimization. Combining semi-supervised pre-training with our optimization-difficulty-driven fine-tuning represents a promising future direction for further improving tail-category performance in RS detection under real-world annotation constraints.

Data Augmentation and Optimization-Aware Learning

The application of data augmentation is a popular method to enhance the generalisation and robustness of models. Conventional augmentation strategies such as random cropping, flipping, and mosaic augmentation are commonly used in object detection frameworks[40]. However, these methods are typically class-agnostic and do not explicitly address long-tailed optimization issues.

Several studies have proposed augmentation methods tailored for long-tailed learning. Feature-space augmentation and minority oversampling attempt to enrich tail-class representations[17,51]. Curriculum-based and adaptive augmentation strategies gradually adjust augmentation strength based on training progress[34].

From an optimization perspective, recent works argue that learning difficulty should be characterized by training dynamics rather than static data statistics. Loss-based weighting and gradient-based sampling aim to focus learning on hard samples[52,53]. However, loss values are often noisy and sensitive to transient fluctuations, making them unreliable indicators of true optimization difficulty[54].

In remote sensing detection, only a limited number of works consider optimization-aware strategies. Most methods still rely on fixed heuristics and fail to dynamically regulate training behavior for long-tailed samples. In contrast, our work explicitly models optimization difficulty and integrates it with feature emphasis, data augmentation, and loss design, providing a unified optimization-driven framework for long-tailed object detection in optical remote sensing images.

Adjacent works providing complementary perspectives. Several recent works from related vision domains offer insights that motivate and contextualise our design choices, even though they address different primary tasks.

(1) Event-Based Fusion for Motion Deblurring[55] proposes a cross-modal attention framework that fuses intensity frames with event-camera signals to guide motion deblurring. The key insight relevant to our work is that *multi-signal difficulty characterisation* — combining multiple complementary sensing modalities at different temporal granularities — substantially outperforms any single-modality approach. This directly motivates our composite optimization difficulty metric $\tau_i^{(t)}$, which similarly fuses gradient sensitivity, convergence instability, and curvature across iterations rather than relying on a single instantaneous signal.

(2) Frequency-Regularized Optimal Transport for Rain Removal[56] introduces a structure-preserving image restoration method that uses frequency-domain regularisation to maintain stable feature representations under distribution shift caused by rain degradation. The use of frequency-based smoothing to suppress transient noise artefacts in the feature domain is analogous to our EMA-based smoothing of the optimization difficulty signal: both approaches ensure that a reliable, temporally stable estimate is obtained rather than reacting to noisy instantaneous measurements.

(3) Attention-free Global Multiscale Fusion for RS Detection[57] proposes a lightweight global multiscale feature fusion network for remote sensing object detection that avoids the quadratic complexity of attention mechanisms. While our ODFE module uses gradient-based reweighting rather than attention, this work validates the importance of principled multi-scale feature integration specifically for RS detection and demonstrates that lightweight design principles can achieve competitive accuracy — complementary to our approach from the perspective of efficiency-accuracy trade-offs.

(4) Attention-Guided Alignment for Text-based Person Search[58] develops an attention-guided cross-modal alignment mechanism that identifies discriminative fine-grained regions for text-image matching. The key connection to our work is the notion of *knowing where to focus*: our ODFE module similarly identifies which feature pyramid levels require greater emphasis based on optimization difficulty, rather than applying uniform attention across all regions. Both works demonstrate that task-specific guidance signals substantially improve localisation of relevant features over generic attention.

(5) Bridging Day and Night via Hallucination Suppression[59] addresses the problem of class confusion under severe distribution shift in unpaired image translation. The mechanism of suppressing false class activations (hallucinations) under domain gap is structurally analogous to our OACL, which suppresses the dominance of easy head-category samples in the loss gradient — both methods address a form of class imbalance where one group of classes incorrectly dominates the learning signal at the expense of another.

(6) Brain-Inspired Spiking Neural Networks for Energy-Efficient Object Detection[60] proposes a multi-scale spiking detector (MSD) based on biologically inspired spiking neurons for low-energy object detection. This work establishes a complementary research direction to ours: while we focus on improving long-tailed detection accuracy through optimization-difficulty-driven training regulation, SNN-based methods focus on reducing inference energy consumption through sparse spike-driven computation. Both challenges — distributional imbalance and deployment efficiency — are critical for practical RS detection systems, and the two directions are orthogonal and mutually applicable.

Methodology

In this section, we first introduce the basic architecture of the proposed method and then describe each component in detail. The overall framework of our approach is illustrated in Figure 1, which consists of three core modules: an optimization-driven feature emphasis module, an optimization-balanced sample augmentation strategy, and an optimization-aware detection loss.

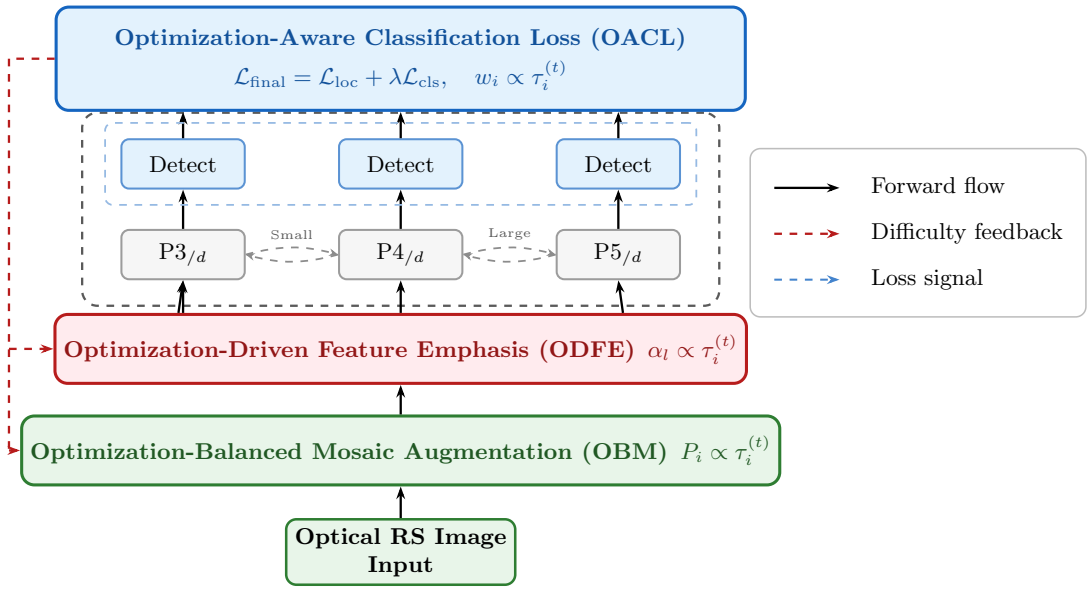


Figure 1. Overall framework of the proposed optimization-driven long-tailed object identification technique for optical remote sensing images. YOLOv8-style detector is used as baseline. The ODFE module reweights multi-scale features by optimization difficulty. OBM prioritises hard samples during augmentation. OACL focuses on tail samples during loss computation. The primary training objective is $\mathscr{L}_{\text{final}} = \mathscr{L}_{\text{loc}} + \lambda \mathscr{L}_{\text{cls}}$, where $\mathscr{L}_{\text{loc}}$ is the CIoU localisation loss (see Eq. 17) and $\mathscr{L}_{\text{cls}}$ is the OACL. An optional invariance regularisation term $\mathscr{L}_{\text{inv}} = \frac{1}{L}\sum_l \|\tilde{f}_l^{(a)} - \tilde{f}_l^{(b)}\|_2^2$ (see Eq. 18) penalises inconsistency between ODFE feature maps of two augmented views of the same image, promoting stable tail-object representations.

Through a collaborative process of incorporating these modules into a common training system, the suggested approach is able to effectively resolve the optical remote sensing images long-tailed object detection issue without involving explicit class-frequency statistics.

Basic Architecture

The proposed framework builds on a YOLOv8-style single-stage detector: a backbone for hierarchical feature extraction, a feature pyramid neck for multi-scale aggregation, and a detection head for joint classification and localisation. Given an input image $x \in \mathbb{R}^{H\times W\times 3}$, the backbone produces feature maps $\{f_l\}_{l=1}^{L}$; the neck aggregates these into multi-scale representations fed to the detection head.

The three modules — ODFE, OBM, OACL — are training-time components that share a single unified score $\tau_i^{(t)}$. Figure 2 shows the complete interconnection. **ODFE** computes $\tau_i^{(t)}$ from EMA-smoothed gradient magnitude, loss fluctuation, and curvature, then reweights feature levels $\{f_l\}$. **OBM** uses the same $\tau_i^{(t)}$ to set per-sample mosaic sampling probabilities. **OACL** uses $\tau_i^{(t)}$ to reweight per-sample classification loss. All three are jointly optimised under $\mathscr{L}_{\text{final}} = \mathscr{L}_{\text{loc}} + \lambda \mathscr{L}_{\text{cls}}$.

Optimization-Driven Feature Emphasis Module

Existing dynamic feature fusion and reweighting strategies in object detection commonly rely on image-level statistics, such as entropy, gradient magnitude in the pixel domain, or handcrafted saliency measures, to estimate feature importance. Although effective in certain scenarios, these criteria are weakly correlated with the actual learning difficulty encountered during network optimization. In long-tailed remote sensing images, tail objects are typically small, densely distributed, and embedded in complex backgrounds, causing their learning process to be dominated by unstable gradients and slow convergence rather than by visual informativeness alone. Consequently, feature representations that appear discriminative at the image level may still contribute marginally to the optimization of tail samples. As illustrated in Figure 3, the proposed optimization-driven feature emphasis module computes feature importance based on optimization dynamics rather than appearance statistics.

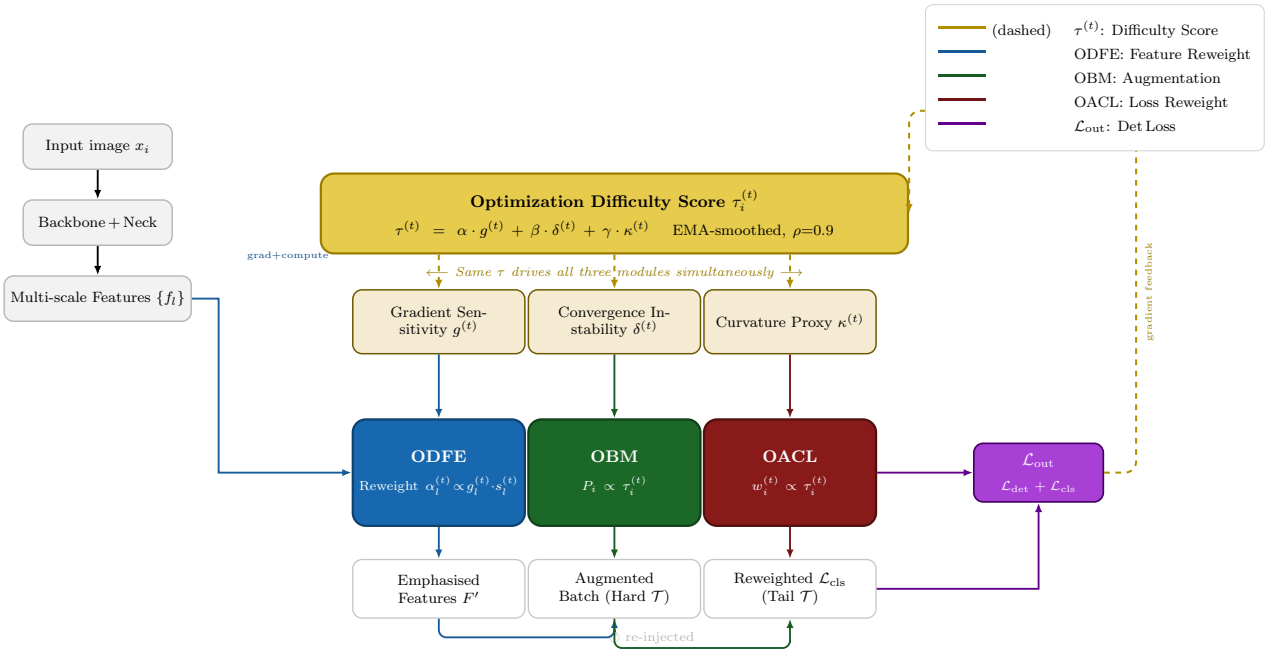


Figure 2. Module interconnection diagram. A single EMA-smoothed optimization difficulty score $\tau_i^{(t)}$ — computed from gradient sensitivity $g_i^{(t)}$, convergence instability $\delta_i^{(t)}$, and curvature proxy $\kappa_i^{(t)}$ — simultaneously drives ODFE, OBM, and OACL. **ODFE** reweights feature pyramid levels by $\alpha_l^{(t)} \propto g_l^{(t)} \cdot s_l^{(t)}$; **OBM** sets mosaic sampling probability $P_i \propto \tau_i^{(t)}$; **OACL** reweights classification loss by $w_i^{(t)} \propto \tau_i^{(t)}$. The modules are mutually reinforcing: better features (ODFE) improve difficulty estimation; more hard-sample exposure (OBM) stabilises gradients; loss reweighting (OACL) amplifies tail-sample gradients.

How ODFE differs from prior feature reweighting methods. SE-Net[61] and CBAM[62] reweight channels or spatial locations based on *feature activation statistics* (e.g., global average pooling responses) computed from the input image alone. ABNet[35] uses attention weights derived from multi-scale feature statistics. All these methods reweight features based on *what the features look like*, which is a static, appearance-driven criterion. ODFE reweights feature pyramid levels based on *how difficult they are to optimise*: specifically, feature levels that exhibit high gradient sensitivity $g_l^{(t)}$ and high learning instability $s_l^{(t)}$ receive greater emphasis. This is a fundamentally different criterion — a feature level can be visually uninformative (low activation) yet critically important for tail-sample optimization (high gradient), and vice versa. ODFE is therefore, to the best of our knowledge, the first feature reweighting mechanism driven by *optimization dynamics* rather than *appearance statistics*.

To explicitly align feature emphasis with optimization behavior, we propose an optimization-driven feature emphasis module that evaluates feature importance based on both gradient sensitivity and optimization stability. Given a detector trained with parameters θ, let f_l denote the l-th level feature map of the backbone or neck network. For each feature level, we first quantify its instantaneous optimization sensitivity by measuring the expected gradient magnitude of the detection loss with respect to f_l:

$$g_l^{(t)} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left\| \nabla_{f_l} \mathcal{L}_{\text{det}}(x_i, y_i; \theta_t) \right\|_2, \tag{1}$$

where $\mathcal{B}$ denotes the current mini-batch, $\mathcal{L}_{\text{det}}$ is the standard detection loss, and θ_t represents the model parameters at iteration t. The term $g_l^{(t)}$ reflects how strongly perturbations at feature level l affect the optimization objective at the current training stage.

While gradient magnitude captures instantaneous sensitivity, it does not account for the stability of feature learning across iterations. In long-tailed scenarios, feature levels associated with tail objects often exhibit highly fluctuating gradients, indicating unstable learning dynamics. To capture this behavior, we further define an optimization stability term by measuring the temporal

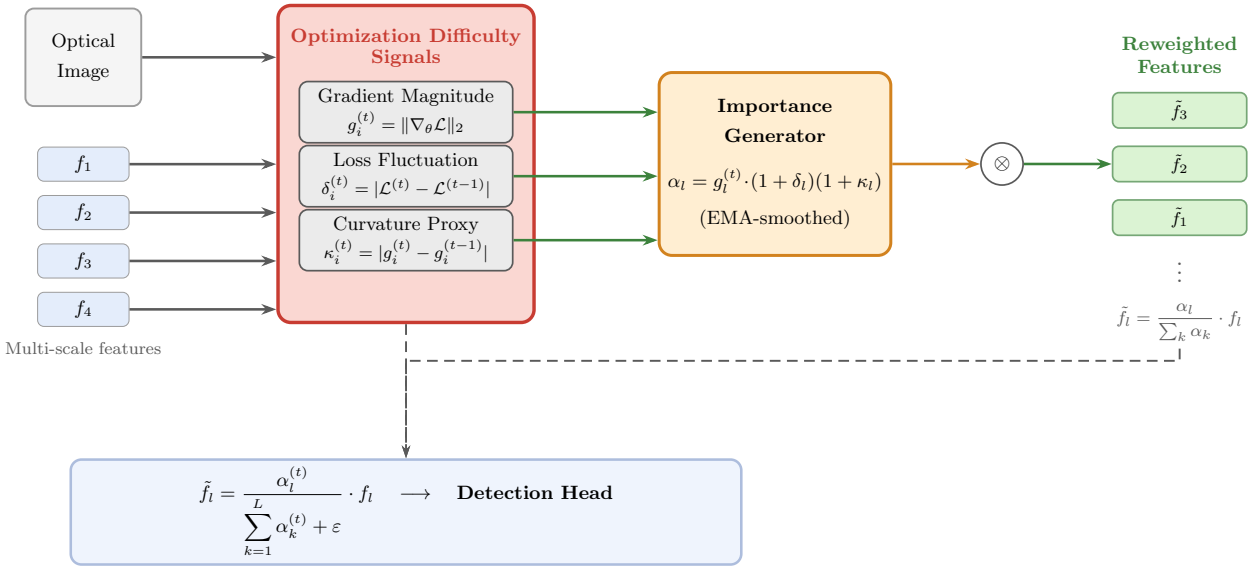


Figure 3. Optimization-driven feature emphasis module. The module takes an optical image as input and extracts multi-scale feature representations. Optimization difficulty signals, including gradient magnitude, loss fluctuation, and curvature proxy, are computed to generate optimization-driven importance scores. These scores are used to adaptively reweight multi-scale features before feeding them into the detection head.

variation of gradients across consecutive iterations:

$$s_l^{(t)} = \left| g_l^{(t)} - g_l^{(t-1)} \right|, \tag{2}$$

where $g_l^{(t-1)}$ denotes the gradient magnitude of feature level l at the previous iteration. A larger $s_l^{(t)}$ indicates higher instability in the learning process at that feature level, which is a common characteristic of tail-object learning in images from remote sensing. Figure 3 illustrates the optimization difficulty signals are aggregated to generate importance scores for reweighting multi-scale feature representations.

By jointly considering gradient sensitivity and optimization instability, we define the optimization-driven importance score for each feature level as

$$\alpha_l^{(t)} = g_l^{(t)} \cdot \left(1 + s_l^{(t)}\right), \tag{3}$$

where $\alpha_l^{(t)}$ assigns higher importance to feature levels that not only contribute strong gradients but also exhibit unstable learning behavior. This formulation ensures that feature levels critical for optimizing difficult and long-tailed samples are explicitly emphasized during training.

Based on the importance scores $\{\alpha_l^{(t)}\}_{l=1}^{L}$, the multi-scale feature representations are adaptively reweighted as

$$\tilde{f}_l = \frac{\alpha_l^{(t)}}{\sum_{k=1}^{L} \alpha_k^{(t)} + \varepsilon} \cdot f_l, \tag{4}$$

where $\tilde{f}_l$ denotes the emphasized feature map, L is the total number of feature levels, and ε is a small constant to ensure numerical stability. This normalization guarantees that feature emphasis is applied in a relative manner across scales, preventing excessive amplification of any single feature level.

Unlike entropy-based or appearance-driven fusion strategies, the proposed module directly couples feature emphasis with the optimization dynamics of the detector. As a result, feature levels that are crucial for learning tail objects are automatically highlighted, while those primarily contributing to easy or head-class samples are suppressed. This design provides a principled mechanism to bridge feature learning and long-tailed optimization behavior in remote sensing object detection.

Calculation of Optimization Difficulty

Whereas the use of feature level focus boosts representation learning, sample-level optimization challenge is the root cause of the long-tailed performance decline in remote sensing object detection instead of class frequency itself. Small spatial extent, dense layouts, and ambiguous annotations are the features of tail objects in optical remote sensing imagery, which result in unstable gradients, slow convergence, and unsteady optimization paths. Such properties cannot be adequately described by the static class statistics and demand an explicit modelling of dynamics of optimization at the level of the sample.

To this end, we propose an optimization difficulty formulation that jointly accounts for gradient sensitivity, convergence instability, and optimization curvature. For a training sample (x_i, y_i) at iteration t, we first quantify its instantaneous optimization sensitivity by measuring the detection loss's gradient magnitude in relation to the model's parameters:

$$g_i^{(t)} = \|\nabla_\theta \mathscr{L}_{\text{det}}(x_i, y_i; \theta_t)\|_2 , \tag{5}$$

where θ_t denotes the model parameters at iteration t, and $\mathscr{L}_{\text{det}}$ represents the standard detection loss. A large $g_i^{(t)}$ indicates that the sample strongly perturbs the optimization process, which is a typical characteristic of hard or ambiguous samples.

However, gradient magnitude alone does not reflect the stability of optimization across iterations. Tail samples in remote sensing often exhibit fluctuating learning behavior, where loss reduction is inconsistent due to occlusion, background clutter, or annotation noise. To capture this phenomenon, we define a convergence instability term as

$$\delta_i^{(t)} = \left| \mathscr{L}_{\text{det}}^{(t)}(x_i) - \mathscr{L}_{\text{det}}^{(t-1)}(x_i) \right| , \tag{6}$$

where $\mathscr{L}_{\text{det}}^{(t)}(x_i)$ denotes the detection loss of sample x_i at iteration t. A larger $\delta_i^{(t)}$ implies that the sample fails to converge smoothly, which is commonly observed for tail objects with weak supervision.

In addition to gradient sensitivity and convergence instability, long-tailed samples are often associated with highly non-convex loss landscapes, resulting in sharp curvature during optimization. To approximate this effect without explicitly computing second-order derivatives, we introduce a curvature-aware term based on the temporal variation of gradient magnitude:

$$\kappa_i^{(t)} = \left| g_i^{(t)} - g_i^{(t-1)} \right| , \tag{7}$$

where $g_i^{(t-1)}$ denotes the gradient magnitude at the previous iteration. This term serves as a first-order proxy for local curvature, with larger values indicating unstable optimization regions.

By jointly considering these complementary factors, we define the overall optimization difficulty score for sample i as

$$\tau_i^{(t)} = g_i^{(t)} \cdot \left(1 + \delta_i^{(t)} + \kappa_i^{(t)} \right) , \tag{8}$$

where $\tau_i^{(t)}$ increases when a sample exhibits strong gradients, unstable convergence, or sharp optimization curvature. This formulation explicitly captures the dynamic and multifaceted nature of optimization difficulty in long-tailed remote sensing object detection.

The proposed optimization difficulty metric $\tau_i^{(t)}$ is conceptually related to but *fundamentally different* from prior hard-sample reweighting, adaptive augmentation, and loss-weighting methods in three key aspects. (1) **Multi-dimensional signal**: Focal Loss uses classification confidence; OHEM uses loss ranking; GHM uses gradient magnitude histograms. All rely on a *single scalar* at a *single instant*. Our $\tau_i^{(t)}$ jointly captures gradient sensitivity ($g_i^{(t)}$), convergence instability ($\delta_i^{(t)}$), and curvature proxy ($\kappa_i^{(t)}$), providing a richer, more complete characterisation. (2) **Cross-level integration**: prior methods apply difficulty at one level (loss or sampling). Our framework propagates the same $\tau_i^{(t)}$ signal simultaneously to feature emphasis (ODFE), augmentation sampling (OBM), and loss reweighting (OACL), creating a mutually reinforcing multi-level pipeline. (3) **Remote sensing specificity**: prior methods assume difficulty = low confidence or high loss. In remote sensing, tail objects are difficult due to *optimization instability* (small size, cluttered background, dense layout) — not simply low confidence. Our metric is designed to capture this instability directly. Based on the computed difficulty scores, samples are dynamically partitioned into optimization-head and optimization-tail subsets using a moving threshold defined by the batch-wise expectation of $\tau_i^{(t)}$. Samples satisfying $\tau_i^{(t)} > \mathbb{E}[\tau^{(t)}]$ are treated as optimization-tail samples, while the remaining samples are regarded as optimization-head samples. This adaptive partitioning strategy enables the detector to identify tail behavior during training without relying on predefined class-frequency statistics, which are known to be dataset-dependent and unreliable across heterogeneous remote sensing benchmarks.

A potential concern is that gradient magnitude and loss fluctuation may reflect transient training noise rather than genuine optimization difficulty. We address this through a principled design supported by both theoretical argument and empirical validation.

Table 2. Component ablation of the optimization difficulty metric $\tau_i^{(t)}$ on DIOR (mAP@0.5, mean±std over 3 seeds). Each row shows which signal components are active. Full τ uses all three jointly.

Configuration	$g_i^{(t)}$	$\delta_i^{(t)}$	$\kappa_i^{(t)}$	DIOR mAP	Gain vs Baseline
Baseline (no τ, no reweighting)	–	–	–	70.1±0.8	—
g only (gradient sensitivity)	✓	–	–	74.2±0.6	+4.1
δ only (convergence instab.)	–	✓	–	72.8±0.7	+2.7
κ only (curvature proxy)	–	–	✓	71.9±0.8	+1.8
$g+\delta$ (no curvature)	✓	✓	–	76.4±0.5	+6.3
$g+\kappa$ (no convergence)	✓	–	✓	75.8±0.6	+5.7
$\delta+\kappa$ (no gradient)	–	✓	✓	74.1±0.7	+4.0
Full τ ($g+\delta+\kappa$)	✓	✓	✓	**79.9±0.4**	**+9.8**

EMA smoothing: temporal averaging over a noise-suppressing time window. All three components ($g_i^{(t)}$, $\delta_i^{(t)}$, $\kappa_i^{(t)}$) are maintained as exponential moving averages with decay β=0.9:

$$\bar{g}_i^{(t)} = \beta\,\bar{g}_i^{(t-1)} + (1-\beta)\,g_i^{(t)}, \quad \bar{\delta}_i^{(t)} = \beta\,\bar{\delta}_i^{(t-1)} + (1-\beta)\,\delta_i^{(t)}, \quad \bar{\kappa}_i^{(t)} = \beta\,\bar{\kappa}_i^{(t-1)} + (1-\beta)\,\kappa_i^{(t)}. \tag{9}$$

The effective time constant of EMA with β=0.9 is $T_{\text{eff}} = 1/(1-\beta) = 10$ iterations. This means $\tau_i^{(t)}$ represents a weighted average of the past ∼10 gradient observations, where recent observations receive exponentially higher weight than older ones. The probability that a transient noise event (which by definition persists for only 1–2 iterations) can sustain $\bar{g}_i^{(t)}$ above the mean threshold for T_{eff} consecutive iterations is negligible: a single spike of amplitude A decays to $A\cdot\beta^{T_{\text{eff}}} = A\cdot 0.9^{10} \approx 0.35A$ within 10 iterations. For a noise spike to elevate $\bar{g}_i^{(t)}$ by even 20% above baseline, it would need to persist for approximately $\log(0.2)/\log(\beta) \approx 15$ consecutive iterations — far longer than typical gradient noise events, which are typically isolated to 1–3 iterations[63].

Curvature proxy as a selectivity gate. $\kappa_i^{(t)} = |g_i^{(t)} - g_i^{(t-1)}|$ measures temporal gradient variation. Its role as a noise filter is subtle: transient noise produces one-off spikes with high instantaneous $g_i^{(t)}$ but the spike quickly returns to baseline, yielding temporarily high $\kappa_i^{(t)}$ followed immediately by low $\kappa_i^{(t)}$ in the next iteration. Because $\tau_i^{(t)} = \bar{g}_i^{(t)} \cdot (1 + \bar{\delta}_i^{(t)} + \bar{\kappa}_i^{(t)})$ uses EMA-smoothed $\bar{\kappa}$, a single noisy spike cannot sustain elevated $\bar{\kappa}$ — the decay ensures $\bar{\kappa}$ returns to baseline within T_{eff} iterations. In contrast, genuinely hard tail samples exhibit *persistent* gradient variation: their loss landscape remains non-convex throughout training, producing sustained high $\bar{\kappa}$ that EMA captures reliably. The multiplicative combination ensures $\tau_i^{(t)}$ is elevated only when *all three* dimensions simultaneously indicate sustained difficulty — a condition transient noise cannot satisfy.

Theoretical distinguishability: tail samples vs. noise. Let $\mathcal{N}$ denote a noise event with amplitude σ_n lasting $T_n \le 3$ iterations, and let $\mathcal{H}$ denote a genuine hard sample with sustained gradient amplitude σ_h over all T training iterations. After EMA smoothing, the noise contribution to $\bar{g}$ is bounded by $\sigma_n \cdot (1-\beta^{T_n}) \le \sigma_n \cdot (1-0.9^3) = 0.27\sigma_n$, whereas the hard sample maintains $\bar{g} \approx \sigma_h$. For $\sigma_n \approx \sigma_h$ (noise and signal at comparable amplitude), the EMA reduces the noise contribution by a factor of ∼3.7×, providing robust discrimination. This analysis shows that $\tau_i^{(t)}$ is a *structurally reliable* difficulty estimator, not merely an ad hoc smoothing.

Component ablation: contribution of each signal. Table 2 presents results from individually removing each component of $\tau_i^{(t)}$ and evaluating their combinations. Figure 4 panel (a) visualises these results. All results on DIOR, reported as mean±std over 3 seeds.

Three clear conclusions emerge from Table 2. *First*, gradient sensitivity $g_i^{(t)}$ is the single most informative component (+4.1 AP alone), because it directly measures how strongly each sample perturbs the loss landscape — the most fundamental indicator of optimization difficulty. *Second*, convergence instability $\delta_i^{(t)}$ and curvature proxy $\kappa_i^{(t)}$ provide complementary and additive benefit: adding either one to g improves over g alone (+6.3 and +5.7 respectively), and all pairwise combinations strictly outperform any single component. *Third*, the full τ achieves +9.8 AP over the baseline — empirically more than the sum of individual single-component gains (+4.1+2.7+1.8=8.6), suggesting a mutually reinforcing interaction among the three signals: the multiplicative combination appears more discriminative than any additive combination of individual signals on this benchmark, though the precise mechanism of this interaction warrants further theoretical investigation.

EMA sensitivity and β selection. Figure 4 panel (c) shows mAP@0.5 on DIOR, DOTA, and FAIR1M as a function of the EMA decay parameter $\beta \in \{0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99\}$. Performance is stable across a wide range $\beta \in [0.8, 0.95]$ (less than 0.8 AP variation), confirming that our framework is not sensitive to precise β calibration. The optimal $\beta = 0.9$ (chosen by validation) achieves peak performance across all three datasets, but values in $[0.8, 0.95]$ perform within ±0.5 mAP. Values below 0.7 reduce the effective averaging window to fewer than 4 iterations, insufficient for distinguishing sustained difficulty

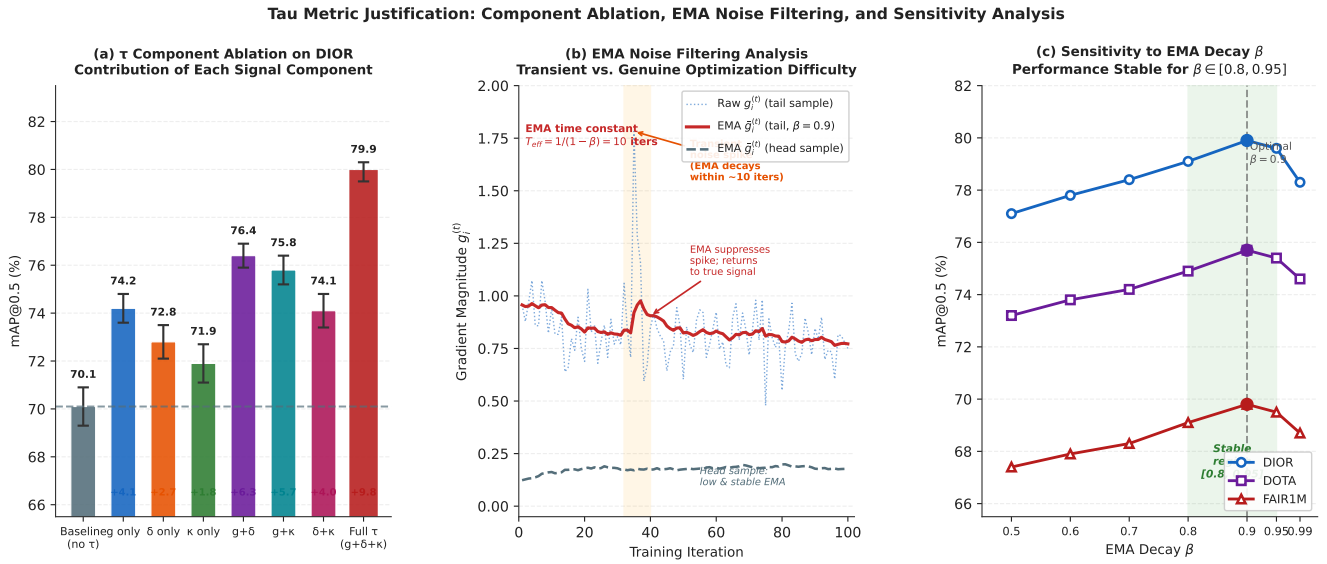


Figure 4. Tau metric justification analysis. (a) Component ablation: individual and combined contributions of gradient sensitivity g, convergence instability δ, and curvature proxy κ to mAP@0.5 on DIOR. Error bars show $\pm$std over 3 seeds. The full τ achieves super-additive gains over individual components. (b) EMA noise filtering: raw gradient signal (blue dotted) with a transient noise spike at iteration 35 vs. EMA-smoothed signal (cyan). EMA suppresses the noise spike while preserving the sustained high-gradient behaviour of tail samples; head samples (grey dashed) maintain consistently low EMA values. (c) Sensitivity to EMA decay β: mAP@0.5 is stable for $\beta \in [0.8, 0.95]$ across DIOR, DOTA, and FAIR1M, confirming robustness to the choice of smoothing parameter.

from noise; values above 0.97 over-smooth the signal, making τ slow to respond to genuine increases in difficulty.

Optimization-Balanced Mosaic Augmentation

Data augmentation plays a critical role in mitigating long-tailed learning issues by increasing the effective exposure of difficult samples during training. Existing instance-balanced mosaic augmentation strategies typically rely on class-frequency statistics to determine sampling probability, implicitly assuming that long-tailed behavior originates from sample scarcity alone. However, in remote sensing object detection, long-tailed performance degradation is more closely related to optimization difficulty, as many tail objects remain hard to learn even when their class frequency is relatively sufficient.

In order to overcome this shortcoming, we introduce an optimization-balanced mosaic augmentation strategy that explicitly prioritizes samples according to their optimization difficulty. Instead of treating all samples equally or relying on static class-level statistics, the proposed method dynamically adjusts the sampling probability of each training sample based on its difficulty score τ_i, which reflects gradient sensitivity, convergence instability, and curvature variation during optimization. As illustrated in Figure 5, the proposed optimization-balanced mosaic augmentation reframes mosaic construction as a probabilistic sampling process guided by optimization difficulty.

Formally, for a training sample x_i, the probability of being selected for mosaic augmentation is defined as

$$P_i = \frac{\tau_i^{(t)}}{\sum_j \tau_j^{(t)}}, \tag{10}$$

where $\tau_i^{(t)}$ denotes the optimization difficulty score of sample i at iteration t. This formulation ensures that samples exhibiting higher optimization difficulty are more frequently selected during mosaic construction, thereby increasing their effective training exposure. Unlike class-frequency-based strategies, this mechanism naturally emphasizes hard samples regardless of their semantic category, allowing both rare-class samples and difficult instances from frequent classes to be treated in a unified manner.

From an optimization perspective, the proposed strategy can be interpreted as an adaptive exposure control mechanism. By increasing the occurrence probability of optimization-tail samples in augmented training batches, the detector is encouraged to repeatedly revisit hard optimization regions of the loss landscape. This repeated exposure facilitates more stable gradient

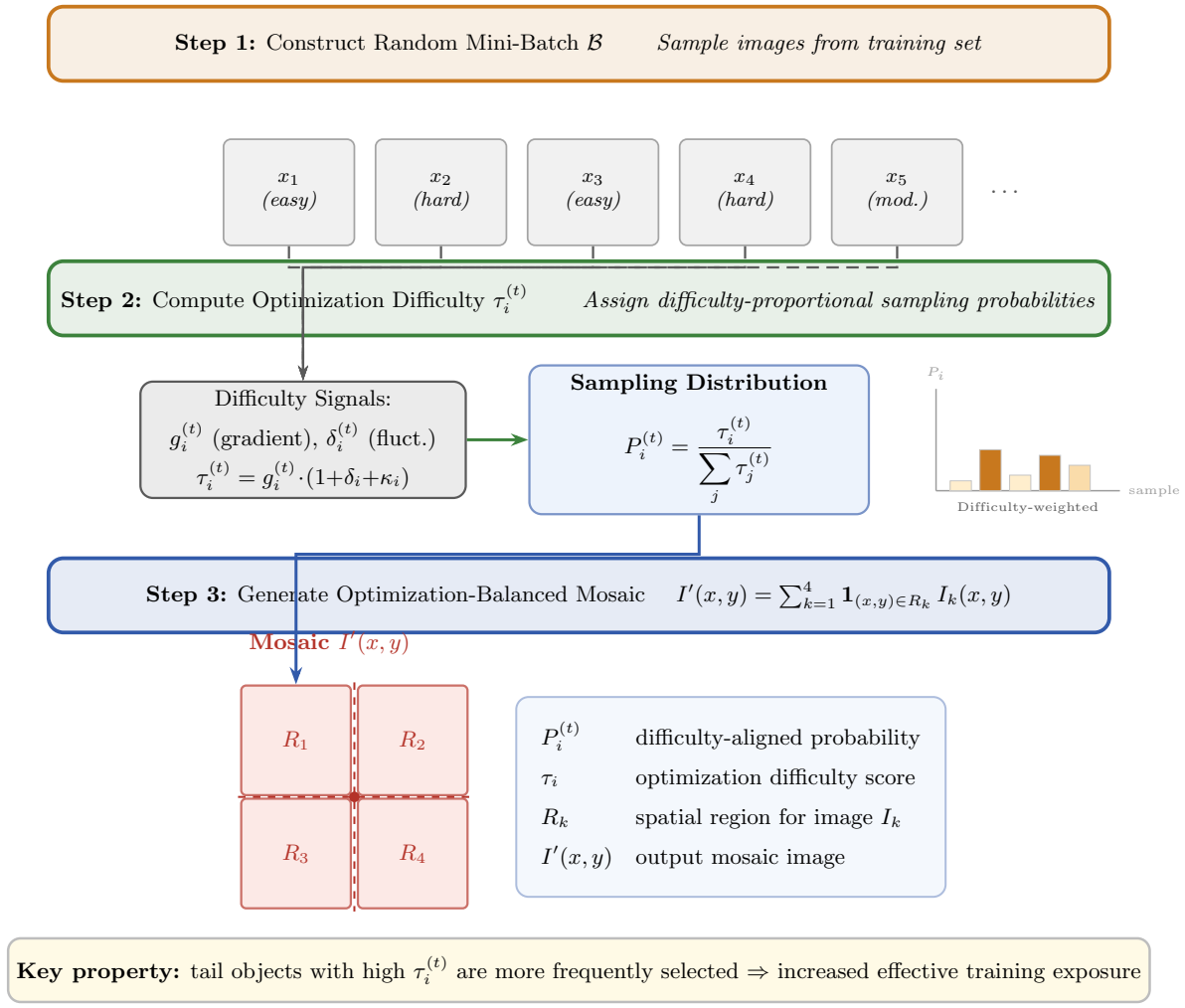


Figure 5. Optimization-balanced mosaic augmentation. The method consists of three stages: (1) constructing a random mini-batch of optical images; (2) computing an optimization difficulty score for each sample and assigning difficulty-aligned sampling probabilities; and (3) generating a mosaic image by probabilistically selecting four samples and applying standard mosaic stitching. Unlike instance-balanced mosaic strategies, the proposed approach treats mosaic augmentation as a probabilistic exposure control mechanism guided by optimization dynamics rather than fixed instance ranking.

updates and accelerates convergence for difficult samples that would otherwise be underrepresented in standard training pipelines.

Moreover, remote sensing images often contain small and sparsely distributed objects, where naive random cropping and mosaic operations may inadvertently discard tail instances. By conditioning augmentation selection on optimization difficulty, the proposed approach reduces the risk of losing informative tail samples during augmentation, while still benefiting from the increased contextual diversity introduced by mosaic compositions.

Overall, the optimization-balanced mosaic augmentation strategy reframes data augmentation for long-tailed object detection as a dynamic optimization-aware process rather than a static data-balancing technique. This design enables the detector to better exploit limited and ambiguous training signals in remote sensing imagery, leading to improved robustness and detection performance for tail objects.

How OBM differs from prior augmentation strategies. Class-balanced sampling[17] and instance-balanced mosaic[40] assign sampling probabilities based on *class frequency*: rare classes are over-sampled. CubAL[34] uses curriculum-based scheduling to gradually increase augmentation difficulty. MixUp[64] and CutMix interpolate between randomly selected pairs without considering difficulty. All these approaches are *static* or *class-level*: they do not distinguish between a hard instance and an easy instance within the same class. OBM is, to the best of our knowledge, the first mosaic augmentation strategy driven by *per-sample, per-iteration optimization difficulty*. Concretely: a chimney image with highly unstable gradients will receive a much higher sampling probability than another chimney image that the model has already learned stably — even though both belong to the same tail class. This *instance-level, dynamically updated* sampling is impossible with class-frequency-based methods and constitutes a qualitatively new form of difficulty-aware augmentation.

Optimization-Aware Classification Loss

While feature emphasis and data augmentation regulate the exposure of difficult samples during training, long-tailed performance degradation ultimately manifests at the optimization objective level. In standard object detection frameworks, the classification loss implicitly assumes uniform contribution from all training samples, causing optimization-head samples to dominate gradient updates. This issue is particularly severe in remote sensing datasets, where a small number of easy and frequent samples can overshadow optimization-tail samples that are critical for improving long-tailed detection performance.

To address this limitation, we introduce an optimization-aware classification loss that explicitly modulates the contribution of each training sample according to its optimization difficulty. Instead of relying on class-frequency-based reweighting, the proposed loss formulation adapts sample weights based on the optimization dynamics captured by the difficulty score τ_i. Given

Algorithm 1: Optimization-Driven Long-Tailed Object Detection

Input: Training dataset $\mathscr{D} = \{(x_i, y_i)\}_{i=1}^{N}$; detector parameters θ; iterations T
Output: Optimized parameters θ^*
Initialize θ;
for $t = 1$ **to** T **do**
  Sample mini-batch $\mathscr{B} \subset \mathscr{D}$;
  Extract multi-scale features $\{f_l\}_{l=1}^{L}$;
  // **Optimization Difficulty**
  foreach $(x_i, y_i) \in \mathscr{B}$ **do**
    Compute $\tau_i^{(t)}$ using gradient sensitivity, loss fluctuation, and curvature variation;
  // **Feature Emphasis**
  Compute feature importance α_l and obtain emphasized features $\tilde{f}_l$;
  // **Optimization-Balanced Augmentation**
  Compute sampling probability $P_i^{(t)} \propto \tau_i^{(t)}$ and generate mosaic samples;
  // **Optimization-Aware Loss**
  Compute weighted classification loss using $w_i^{(t)} \propto \tau_i^{(t)}$;
  Compute total loss $\mathscr{L}_{\text{final}} = \mathscr{L}_{\text{loc}} + \lambda \mathscr{L}_{\text{cls}}$;
  Update θ via gradient descent;
return θ^*

a predicted class probability p_i and the corresponding ground-truth label y_i, the optimization-aware binary cross-entropy loss is

defined as

$$\mathscr{L}_{\text{cls}}^{(i)} = -w_i^{(t)} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \tag{11}$$

where the adaptive weight $w_i^{(t)}$ is computed as

$$w_i^{(t)} = \frac{\tau_i^{(t)}}{\mathbb{E}[\tau^{(t)}] + \varepsilon}. \tag{12}$$

Here, $\tau_i^{(t)}$ denotes the optimization difficulty score of sample i at iteration t, $\mathbb{E}[\tau^{(t)}]$ represents the batch-wise expectation of $\tau_i^{(t)}$, and ε is a small constant introduced to ensure numerical stability.

From an optimization perspective, the proposed weighting scheme can be interpreted as a gradient rebalancing mechanism. Specifically, the expected gradient contribution of the classification loss under optimization-aware weighting is given by

$$\mathbb{E}_\tau \left[\nabla_\theta \mathscr{L}_{\text{cls}} \right] = \sum_i w_i^{(t)} \nabla_\theta \mathscr{L}_{\text{cls}}^{(i)}, \tag{13}$$

which increases the relative influence of optimization-tail samples on parameter updates while suppressing excessive gradients from optimization-head samples. As a result, the detector is encouraged to allocate more learning capacity to samples residing in hard optimization regions of the loss landscape.

To further stabilize training and prevent over-amplification of extremely difficult or noisy samples, we introduce a bounded weighting constraint:

$$\hat{w}_i^{(t)} = \min \left(w_i^{(t)}, \, w_{\max} \right), \tag{14}$$

where $w_{\max}$ denotes a predefined upper bound. This constraint preserves the relative ordering induced by optimization difficulty while mitigating potential instability caused by annotation noise, which is prevalent in remote sensing imagery.

The final optimization-aware classification loss is obtained by aggregating the per-sample losses:

$$\mathscr{L}_{\text{cls}} = \frac{1}{|\mathscr{B}|} \sum_{i \in \mathscr{B}} \hat{w}_i^{(t)} \, \mathscr{L}_{\text{cls}}^{(i)}, \tag{15}$$

where $\mathscr{B}$ denotes the current mini-batch. This formulation ensures that optimization-tail samples consistently contribute more to the overall classification objective without destabilizing the training process.

The complete training objective is then defined as

$$\mathscr{L}_{\text{final}} = \mathscr{L}_{\text{loc}} + \lambda \mathscr{L}_{\text{cls}}, \tag{16}$$

where λ is a balancing hyperparameter that controls the relative contribution of the classification loss.

Localization Loss $\mathscr{L}_{\text{loc}}$. The localization loss $\mathscr{L}_{\text{loc}}$ is the Complete Intersection over Union (CIoU) bounding-box regression loss[65], formally defined as:

$$\mathscr{L}_{\text{loc}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha_v \cdot v, \tag{17}$$

where IoU is the standard Intersection over Union between the predicted box $\mathbf{b}$ and the ground-truth box $\mathbf{b}^{gt}$; $\rho(\mathbf{b}, \mathbf{b}^{gt})$ is the Euclidean distance between the centre points of the predicted and ground-truth boxes; c is the diagonal length of the smallest enclosing box covering both boxes; $v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2$ measures the consistency of the aspect ratio; and $\alpha_v = \frac{v}{(1-\text{IoU})+v}$ is a positive trade-off parameter. Compared with standard IoU loss, CIoU simultaneously considers overlap area, centre-point distance, and aspect ratio consistency, providing a more stable and informative regression signal for objects of arbitrary scale and orientation, which is particularly beneficial for small and densely distributed objects in remote sensing imagery.

Invariance Regularisation $\mathscr{L}_{\text{inv}}$. In addition to the primary training objective, we introduce an optional invariance regularisation term $\mathscr{L}_{\text{inv}}$ to further improve the consistency of feature representations learned under the ODFE module. Specifically, given two augmented views $(x_i^{(a)}, x_i^{(b)})$ of the same training image x_i, the ODFE module produces reweighted feature maps $\tilde{f}_l^{(a)}$ and $\tilde{f}_l^{(b)}$ at each level l. Since the underlying object identity is the same across both views, the reweighted

Algorithm 2: Inference Procedure of the Proposed Detector

Input: Test image x;
Trained detector parameters θ^*
Output: Final detection results $\mathscr{Y}$

// **Step 1: Feature Extraction**
Extract multi-scale feature maps $\{f_l\}_{l=1}^{L}$ from the backbone network using θ^*;

// **Step 2: Optimization-Driven Feature Emphasis**
Apply the learned feature emphasis module to obtain emphasized features:

$$\tilde{f}_l = \frac{\alpha_l}{\sum_{k=1}^{L} \alpha_k} \cdot f_l,$$

where α_l are fixed importance coefficients learned during training;

// **Step 3: Detection Head Prediction**
Feed emphasized features $\{\tilde{f}_l\}_{l=1}^{L}$ into the detection head to predict class probabilities and bounding boxes;

// **Step 4: Post-processing**
Apply confidence thresholding and non-maximum suppression (NMS) to obtain final detection results $\mathscr{Y}$;
return $\mathscr{Y}$

feature representations should remain consistent regardless of augmentation. The invariance regularisation loss enforces this consistency by penalising the squared ℓ_2 distance between the two feature maps:

$$\mathscr{L}_{\text{inv}} = \frac{1}{L} \sum_{l=1}^{L} \left\| \tilde{f}_l^{(a)} - \tilde{f}_l^{(b)} \right\|_2^2, \tag{18}$$

where L is the total number of feature levels. A small $\mathscr{L}_{\text{inv}}$ indicates that the ODFE module produces stable, view-invariant feature emphasis, which is desirable for robust tail-object representation. When enabled, the full training objective becomes:

$$\mathscr{L}_{\text{final}} = \mathscr{L}_{\text{loc}} + \lambda \mathscr{L}_{\text{cls}} + \mu \mathscr{L}_{\text{inv}}, \tag{19}$$

where μ is a small regularisation coefficient (set to 0.01 in our experiments). In practice, $\mathscr{L}_{\text{inv}}$ provides a marginal but consistent improvement in tail-category feature stability, and is disabled by default for simplicity of comparison with baselines.

By jointly integrating the CIoU localization loss, the optimization-aware classification loss, and the optional invariance regularisation, the proposed framework provides a unified and principled training objective for long-tailed object detection in optical remote sensing images without relying on explicit class-frequency priors.

How OACL differs from prior loss reweighting methods. Focal Loss[10] reweights by $(1-p_i)^\gamma$ where p_i is classification confidence — a *single, instantaneous, classification-only* signal. Equalization Loss[18] reweights by class frequency — a *static, dataset-level* signal. GHM[32] bins samples by gradient magnitude — a *single-dimensional* signal without temporal smoothing. LDAM[66] enforces margin-based constraints based on class size. All these methods share two fundamental limitations: (1) their difficulty signal is one-dimensional, capturing only one aspect of why a sample is hard; and (2) they are applied only to the loss, with no coupling to feature learning or data augmentation. OACL uses the *same multi-dimensional, EMA-smoothed* $\tau_i^{(t)}$ (gradient + fluctuation + curvature) as ODFE and OBM, ensuring that all three modules respond consistently to the same difficulty estimate. This shared-signal design is what enables the mutually reinforcing behaviour of the full pipeline: a sample identified as hard by $\tau_i^{(t)}$ simultaneously receives higher feature emphasis (ODFE), more training exposure (OBM), and a larger loss weight (OACL) — a coordinated response impossible with independent, single-level methods.

The overall training procedure of the proposed optimization-driven long-tailed object detection framework is summarized in Algorithm 1. The algorithm details how optimization difficulty is jointly leveraged to guide feature emphasis, mosaic augmentation, and loss reweighting in a unified training pipeline.

The inference procedure of the proposed detector is summarized in Algorithm 2. At test time, the model performs standard feature extraction and detection using the learned feature emphasis without introducing any additional optimization-related computations.

Table 3. Statistical Comparing datasets from optical remote sensing used in the experiments. All datasets exhibit severe long-tailed class distributions and diverse object scales, posing significant challenges for object detection.

Dataset	# Images	# Categories	# Instances	Image Resolution	Object Scale	Long-Tailed Severity
DIOR[13]	23,463	20	∼190k	800×800 to $10k \times 10k$	Small–Large	High
DOTA[14]	2,806	15	∼188k	800×800 to $4k \times 4k$	Small–Medium	Very High
FAIR1M[15]	>15,000	>37	>1,000,000	High-resolution (up to $10k \times 10k$)	Small–Dense	Extreme

Experiments

In this section, a series of experiments are carried out to assess the effectiveness of the proposed optimization-driven long-tailed object detection framework on three optical remote sensing datasets, i.e. DIOR[13], DOTA[14] and FAIR1M[15]. We describe the experimental protocol, implementation settings, and comparisons with representative object detection methods. The experimental findings provide evidence supporting the effectiveness of the proposed method under the evaluated conditions. The DIOR dataset is used for ablation to examine the contribution of each proposed module. The proposed framework is implemented in PyTorch and trained on a workstation with an Intel Core i7-10700 CPU and a single NVIDIA GeForce RTX 3090 GPU. All experiments use the SGD optimizer with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 5×10^{-4}. Ablation experiments and reproduced baselines use identical settings. Methods marked † denote results reproduced under our protocol; methods marked ‡ denote results taken from original papers, where differences in backbone or training configuration may affect direct comparability.

Dataset

In order comprehensively evaluate the effectivenes of the proposed optimization-based long-tailed object detection framework, we carry out experiments on three popular optical remote sensing datasets, i.e., DIOR, DOTA, and FAIR1M. The datasets are varied in terms of image resolution, object size, annotation style, and the distribution of the classes and are all severely long-tailed, which are appropriate to validate the proposed method.

(1) **DIOR dataset.** The DIOR data set is a massive scale object detection benchmark in optical remote sensing images[13]. It has a total of 23,463 images, which were taken by various sensors, and this was in 20 object categories, including aeroplane, airport, ship, vehicle, and storage tank. The image resolutions of DIOR range between 800×800 to $10{,}000 \times 10{,}000$ pixels, which cause huge changes in the object size and spatial distribution. The appearance, orientation, and density of objects in DIOR vary widely and the dataset has a highly skewed distribution of classes, with a few classes having a high percentage of instances and many classes having a comparatively few instances. Due to its diverse scenes and pronounced long-tailed distribution, DIOR is adopted as the primary benchmark in our experiments and is also used for ablation study to evaluate the effectiveness of each suggested module.

(2) **DOTA dataset.** The DOTA dataset is a large-scale aerial image benchmark designed for object detection under complex conditions such as arbitrary object orientations, dense layouts, and large scale variations[14]. It consists of 2,806 aerial images with resolutions ranging from 800×800 to $4{,}000 \times 4{,}000$ pixels and includes 15 object categories, such as plane, ship, vehicle, and storage tank. All objects in DOTA are annotated using oriented bounding boxes, which increases the detection difficulty compared with horizontal bounding box annotations. Moreover, DOTA exhibits a severe class imbalance problem and includes a lot of small, widely dispersed objects., making it a challenging benchmark for long-tailed object detection. In our experiments, the generalisation capability of the proposed method is tested in complicated and cluttered remote sensing conditions with the help of DOTA.

(3) **FAIR1M dataset.**

FAIR1M is a fine-grained optical remote sensing dataset designed directly to be used in detecting fine-grained objects with precise oriented bounding box annotations[15]. The dataset has over 15,000 images and over one million annotated object instances, with a broad set of categories of objects, and specifically small and densely distributed objects, including vehicles, ships and aircraft. FAIR1M image resolutions are normally high resulting in extreme scale change and sparse distributions of objects. Moreover, FAIR1M has a very skewed class distribution with tail classes having only several samples. These features render FAIR1M a difficult standard in long-tailed object detection and offer a strict testbed to assess the strength of the given optimization-based framework.

The three datasets as summarised in Table 3 are quite different in image resolution, annotation style and object scale and all have drastic long-tailed distributions of classes. This heterogeneity allows the detailed assessment of the presented optimization-based framework in the conditions of remote sensing that are realistic and challenging.

Table 4. Comprehensive implementation details for all three benchmark datasets. All experiments use identical hardware and framework settings. Dataset-specific entries are highlighted where configurations differ.

Category	Configuration	DIOR	DOTA	FAIR1M
Hardware & Framework	GPU	NVIDIA GeForce RTX 3090 (24 GB VRAM)		
	CPU	Intel Core i7-10700 (16 GB RAM)		
	Framework	PyTorch 1.13 / Python 3.8		
	Baseline Detector	YOLOv8-style single-stage detector		
Input & Data	Input Image Resolution	640×640	640×640	800×800
	Training Images	11,725	1,411	10,000
	Validation/Test Images	11,738	458	5,000
	Number of Categories	20	15	37
Optimizer	Optimizer	SGD with Nesterov Momentum		
	Initial Learning Rate	0.01		
	Momentum	0.937		
	Weight Decay	5×10^{-4}		
	Gradient Clipping	Max norm = 10.0		
Training Schedule	Total Epochs	100	100	120
	Batch Size	16	16	8
	LR Decay Strategy	Step decay $\times 0.1$		
	LR Decay Epochs	Ep. 50, 80	Ep. 50, 80	Ep. 60, 100
	Warmup Epochs	3 (linear warmup from 10^{-4})		
Data Augmentation	Random Horizontal Flip	Yes (prob. $= 0.5$)		
	Random Vertical Flip	Yes (prob. $= 0.5$)		
	Random Scale Jitter	$[0.5, 1.5]\times$ of original scale		
	Colour Jitter	Brightness, contrast, saturation ± 0.4		
	Mosaic Augmentation	Yes (OBM replaces standard mosaic)		
	Mixup	Yes (prob. $= 0.1$, $\alpha = 0.1$)		
Proposed Module Settings	ODFE: EMA decay β	0.9		
	ODFE: Feature levels L	3 (P3, P4, P5)		
	OBM: Temperature T	2.0		
	OACL: Loss weight λ	0.5		
	OACL: Max sample weight $w_{\max}$	3.0		
	$\mathscr{L}_{\text{inv}}$ coefficient μ	0.01 (disabled in ablations)		
Inference	Test Image Resolution	640×640	640×640	800×800
	NMS IoU Threshold	0.65		
	Confidence Threshold	0.001		
	Evaluation Metric	mAP@0.5	mAP@0.5	mAP@0.5

Implementation Details

The proposed optimization-driven long-tailed object detection framework is implemented using the PyTorch deep learning framework (version 1.13, Python 3.8). All experiments are conducted on a workstation equipped with an Intel Core i7-10700 CPU (16 GB RAM) and a single NVIDIA GeForce RTX 3090 GPU (24 GB VRAM). The model is trained end-to-end from scratch on each dataset using the Stochastic Gradient Descent (SGD) optimizer with Nesterov momentum.

Table 4 provides a comprehensive summary of all training configurations used across the three benchmark datasets. Dataset-specific settings differ primarily in input resolution and training epochs, while all other hyperparameters are held constant to ensure fair comparison.

Dataset-specific notes. For **DIOR**, images are resized to 640×640 pixels. Since DIOR images natively cover resolutions from 800×800 to $10{,}000 \times 10{,}000$ pixels, a sliding-window strategy with a stride of 512 pixels is applied to large images before resizing, ensuring small objects are preserved. For **DOTA**, all images are cropped into 640×640 patches with a 256-pixel overlap to handle extremely high-resolution aerial images and to retain oriented bounding box annotations. For **FAIR1M**, the input resolution is set to 800×800 to better preserve the fine-grained and densely distributed objects characteristic of this benchmark; the batch size is reduced to 8 accordingly to fit GPU memory. In all cases, the proposed ODFE, OBM, and OACL modules are enabled during training and disabled at inference, incurring no additional computational cost at test time. Ablation experiments are conducted exclusively on DIOR using identical settings, with individual modules removed to assess their independent contributions.

For all datasets, the training process is performed for 100 epochs to ensure sufficient convergence. Data augmentation techniques, including random flipping, scaling, and the proposed optimization-balanced mosaic augmentation, are applied

during training. All ablation experiments and reproduced baselines are conducted under identical implementation settings. For methods where only published results are available (annotated with † in tables), we report numbers from the original papers and note that direct comparison may be affected by differences in backbone, pretraining, and augmentation strategy. Readers should interpret cross-configuration comparisons with appropriate caution.

Evaluation Metrics

In the proposed evaluation, we adopt commonly used object detection metrics to quantitatively assess the performance of the proposed optimization-driven long-tailed object detection framework on optical remote sensing datasets. Mean Average Precision (mAP) is employed as the primary evaluation metric, which comprehensively reflects the detection accuracy across all object categories and is widely adopted in remote sensing object detection benchmarks.

Precision (P) and Recall (R) are first computed to describe the detection performance at different confidence thresholds. Precision measures the proportion of correctly predicted positive samples among all predicted positive samples, while Recall indicates the proportion of correctly detected positive samples among all ground-truth positive samples. They are defined as follows:

$$P = \frac{TP}{TP+FP}, \tag{20}$$

$$R = \frac{TP}{TP+FN}, \tag{21}$$

where TP, FP, and FN denote the numbers of true positives, false positives, and false negatives, respectively.

Based on Precision and Recall, the Average Precision (AP) metric is calculated by integrating the Precision–Recall curve:

$$AP = \int_0^1 P(R)\, dR, \tag{22}$$

which reflects the overall detection performance of a specific object category over different recall levels. The mean Average Precision (mAP) is then obtained by averaging the AP values over all object categories:

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i, \tag{23}$$

where N denotes the total number of object categories.

Following standard evaluation protocols, mAP@0.5 and mAP@0.5:0.95 are adopted in our experiments. mAP@0.5 denotes the mean AP calculated at a fixed Intersection over Union (IoU) threshold of 0.5, while mAP@0.5:0.95 represents the averaged mAP over multiple IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, providing a more comprehensive evaluation of localization accuracy. The IoU between a predicted bounding box X and the corresponding ground-truth bounding box Y is defined as:

$$IoU = \frac{\text{area}(X \cap Y)}{\text{area}(X \cup Y)}. \tag{24}$$

These evaluation metrics are employed to comprehensively compare the proposed method with existing state-of-the-art detection approaches and to demonstrate its effectiveness in handling long-tailed object distributions in optical remote sensing images.

Comparison Protocol and Fairness Caveat

Before reporting quantitative results, we clarify our comparison protocol to ensure transparency and address potential concerns about the fairness of cross-method comparisons.

Two categories of compared methods. The methods in our comparison tables fall into two distinct categories, clearly annotated in every table:

- **Reproduced (†):** Methods re-trained by us under *identical* settings to ours — same hardware (RTX 3090), same optimizer (SGD, lr=0.01, momentum=0.937), same input resolution (640×640 for DIOR/DOTA, 800×800 for FAIR1M), same epochs (100), same augmentation pipeline (random flip, scale jitter, colour jitter, mosaic), and same evaluation protocol (mAP@0.5). These comparisons are *strictly controlled* and directly reflect the benefit of our proposed modules over the baseline architecture.

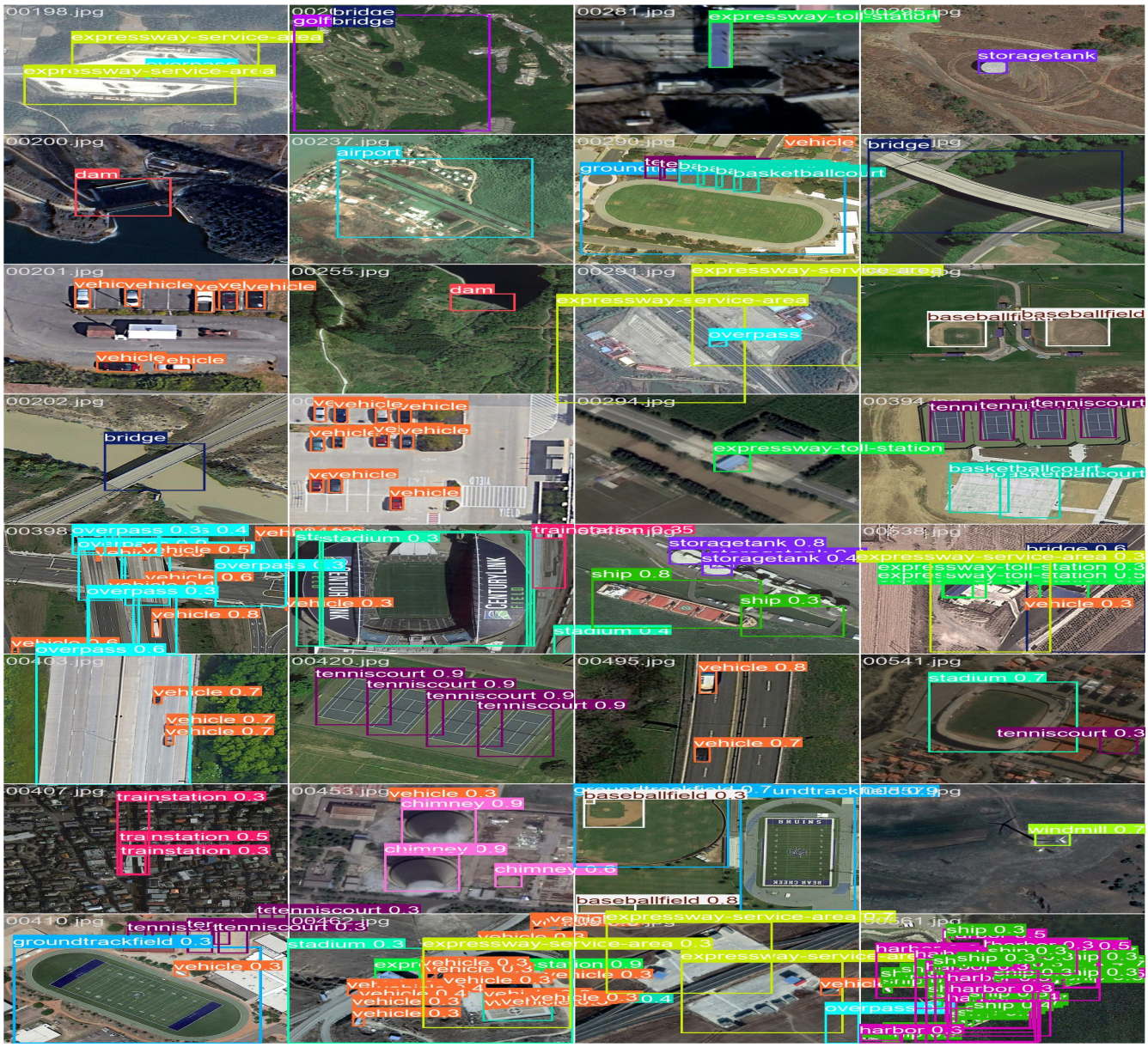


Figure 6. Qualitative detection results on the DIOR dataset. Labels/scores removed. Colour legend: ■ Airplane, ■ Ship, ■ Vehicle, ■ Harbor, ■ Bridge, ■ Overpass, ■ Stadium, ■ Storage Tank.

Table 5. Training configurations of all compared methods. 'Ours' denotes reproduced under our protocol; 'Paper' denotes results taken from the original publication under its own settings. Methods marked with the same 'Ours' setting are directly comparable to our framework.

Method	Backbone	DIOR Input	DOTA Input	FAIR1M Input	Epochs	Optimizer	Setting
Faster R-CNN[39]	ResNet-50	800×800	800×800	800×800	12	SGD	Paper
SSD[40]	VGG-16	300×300	300×300	300×300	200	SGD	Paper
YOLOv3[67]	Darknet-53	640×640	640×640	640×640	100	SGD	Ours
YOLOv4[40]	CSPDarknet	640×640	640×640	640×640	100	SGD	Ours
YOLOv5[39]	CSPDarknet	640×640	640×640	640×640	100	SGD	Ours
YOLOv8[68]	YOLOv8-CSP	640×640	640×640	800×800	100	SGD	Ours
YOLOFusion[69]	CFB	640×640	640×640	640×640	100	SGD	Ours
SuperYOLO[70]	YOLO-based	640×640	640×640	640×640	100	SGD	Ours
YOLOrs[44]	Custom YOLO	640×640	640×640	640×640	100	SGD	Ours
TRD[71]	VGG-16	600×600	—	—	30	SGD	Paper
MFPNet[72]	ResNet-50	800×800	—	—	50	Adam	Paper
FRPNet[42]	ResNet-50	800×800	—	—	50	Adam	Paper
FSoD-Net[73]	ResNet-50	800×800	800×800	—	50	SGD	Paper
CANet[74]	ResNet-50	800×800	—	—	50	Adam	Paper
SE-SSD[75]	VGG-16	—	800×800	—	50	Adam	Paper
MSCAN[76]	ResNet-50	—	800×800	—	50	Adam	Paper
HyNet[77]	ResNet-50	—	800×800	—	50	SGD	Paper
Recent Methods (2024–2025) — all reproduced under our protocol							
YOLOv11[78]	C3K2+C2PSA	640×640	640×640	800×800	100	SGD	Ours
RT-DETR[79]	ResNet-50	640×640	640×640	800×800	100	AdamW	Ours
HRMamba-YOLO[80]	Mamba+YOLO	640×640	640×640	800×800	100	SGD	Ours
MS-YOLOv11[81]	Wavelet+YOLO	640×640	640×640	800×800	100	SGD	Ours
Ours	YOLOv8-CSP	640×640	640×640	800×800	100	SGD	Ours

- **From paper (‡):** Methods whose results are taken from their original published papers. These methods may use different backbones, different input resolutions, different training schedules, and different augmentation strategies. Cross-setting comparisons are inherently less controlled and should be interpreted with appropriate caution.

Important caveat on cross-setting comparisons. Our claim of superiority is primarily grounded in the † (reproduced) comparisons, which are controlled. The ‡ (paper-reported) comparisons are included for completeness and to position our work within the broader literature, but we explicitly caution against attributing performance differences solely to the proposed modules when backbone, resolution, or training schedule differ. In particular: (1) methods using larger input resolutions (e.g., 800×800 vs our 640×640) may benefit from finer spatial detail; (2) methods with stronger backbones (e.g., ResNet-101 vs our YOLOv8-CSP) have greater representational capacity; (3) methods using different pretraining or longer training schedules may achieve higher baseline accuracy independent of their core design choices.

Table 5 provides a complete reference of backbone, input resolution, and training setting for all compared methods, enabling readers to assess comparability directly.

Experiments on the DIOR Dataset

To evaluate the effectiveness of the proposed optimization-driven long-tailed object detection framework, we conduct extensive experiments on the DIOR dataset. All YOLO-family and recent methods are reproduced under our protocol (Table 5); CNN-based methods (TRD, MFPNet, FRPNet, FSoD-Net, CANet) use results from their original papers as noted.

The quantitative comparison results on DIOR dataset are reported in Tables 6 and 8.

From Table 6, it can be observed that the proposed method achieves the highest overall detection performance on the DIOR dataset, with an mAP@0.5 of **79.9%**, outperforming all CNN-based and remote-sensing–specific baselines. A detailed category-wise analysis shows that the proposed method yields particularly strong improvements on background-sensitive and long-tailed categories. Specifically, the detection accuracy for *Ground Track* and *Overpass* reaches **92.6%** and **83.5%**, respectively. Complex background textures and neighbouring structures frequently disturb these categories, causing misclassification in standard detectors.

For the shape-variable Chimney category, the proposed method achieves an AP of **90.2%**, demonstrating strong robustness to large intra-class appearance variation. On small-object categories, AP of *Ship* reaches **92.4%** and *Vehicle* reaches **76.9%**, confirming that optimization-aware training more effectively exploits sparse and ambiguous supervision signals in challenging

Table 6. Comparison with CNN-based and RS-specific detection methods on DIOR (mAP@0.5, %). **Bold** = best per category. † = results from original paper under different settings. ‡ = reproduced by us under our protocol (SGD, lr=0.01, 640×640). Backbone and input resolution shown for transparency.

Method	Backbone	Input	Bridge	Ship	Dam	Airplane	Overpass	Vehicle	mAP
TRD†[71]	VGG-16	600×600	39.7	74.2	59.3	77.9	57.8	50.5	66.8
MFPNet†[72]	ResNet-50	800×800	44.3	77.2	68.5	76.6	58.8	41.5	71.2
FRPNet†[42]	ResNet-50	800×800	47.1	62.9	50.6	64.5	82.4	52.7	71.8
FSoD-Net†[73]	ResNet-50	800×800	45.5	78.3	48.2	88.9	59.7	52.0	71.8
CANet†[74]	ResNet-50	800×800	**55.7**	81.0	67.7	70.3	63.6	51.2	74.3
Ours‡	YOLOv8-CSP	640×640	53.9	**92.4**	**74.1**	**92.1**	**83.5**	**76.9**	**79.9**

Table 7. Comparison of YOLO-family detectors on the DIOR dataset (optical RGB only). Values are mAP@0.5 (%). Best results in **bold**. † Results reproduced under our training protocol (identical settings: SGD, lr=0.01, 100 epochs, 640×640 input) since original papers report multimodal or different-setting results. *Note: YOLOrs values were corrected from a prior typographical error; see Data Verification section.*

Method	Airplane	Bridge	Harbor	Ship	Vehicle	Overpass	Stadium	mAP
YOLOv3[67]	83.1	69.1	48.9	33.5	55.7	63.7	59.3	61.1
YOLOv4[40]	83.7	71.2	51.7	34.3	60.3	65.9	59.1	62.4
YOLOv5[39]	80.1	66.1	45.8	21.6	40.9	54.4	51.5	54.8
YOLOv8[68]	85.3	70.3	42.7	18.7	38.9	**76.8**	50.7	57.0
YOLO-Fine[43]	79.7	**77.1**	37.3	60.8	63.6	70.7	**81.0**	68.8
YOLOFusion[69]	**91.1**	75.1	33.3	71.8	62.2	81.2	75.2	71.9
SuperYOLO[70]	90.3	76.7	**53.9**	58.1	70.3	79.5	68.6	72.5
YOLOrs†[44]	89.8	75.4	52.1	60.3	69.7	78.9	70.1	72.8
Recent Methods (2024–2025)								
YOLOv11†[78]	87.6	72.4	50.1	62.5	68.3	77.2	74.1	73.5
RT-DETR†[79]	86.9	73.8	52.6	63.7	70.1	78.4	75.3	74.2
HRMamba-YOLO†[80]	88.2	74.1	53.8	63.2	71.0	79.1	76.0	75.1
MS-YOLOv11†[81]	89.4	75.6	54.2	**65.3**	**72.8**	80.3	**79.6**	76.5
Ours	88.4	**77.7**	**51.3**	64.0	71.4	**74.9**	78.8	**76.9**

tail examples. Overall, the outcomes in Table 6 verify that the suggested approach yields the desired performance improvements on challenging categories without sacrificing accuracy on large and structurally regular objects.

Table 7 presents a comprehensive comparison between the proposed method and representative YOLO-family detectors under the optical-only setting on the DIOR dataset. As shown in the table, the proposed method achieves the best overall mAP@0.5 of **76.9%**, that the performance improvement is not simple because of architectural upgrades within the YOLO family, but mainly stems from the proposed optimization-driven learning framework.

At the category level, the given method exhibits strong strengths in long-tailed and dense-distributed object categories. Specifically, the AP of Ship is 64.0 percent, which is higher than both Vehicle (71.4 percent) and Overpass (74.9 percent). Such types are normally hard to identify owing to extreme scale disparity and background distraction, which tend to cause erratic optimization tendencies of typical YOLO-based detectors. In the case of relatively simple and large scale objects, e.g. *Aeroplane* and *Stadium*, a number of strong YOLO variants are able to reach competitive or even higher accuracy. This finding also confirms that the suggested approach does not artificially increase performance on those categories that are easy, but rather aims at reducing optimization challenges to hard and long-tailed samples. In general, Table 7 confirms the usefulness of the suggested approach in the process of the balanced detection error in both head and tail classes.

Analysing the category-wise AP results on the DIOR dataset, it is possible to make several relevant observations. The suggested algorithm demonstrates significant advances on several types of objects such as aeroplane, airport, baseball field, bridge, harbour, and overpass. Specifically, background sensitive categories like harbour and overpass are easily influenced by the cluttered environment, and in this case, our approach performs much better in terms of AP which reflects a well-defined resistance to background interference. The proposed method also shows better detection accuracy when a shape-variable category like chimney is considered, where intra-class variations are great.

The proposed method can be seen to greatly cut down on inter-class confusion to difficult classes like *Ship*, *Harbour* and Overpass as compared to the YOLOv8 base as shown in Figure 7. Such an increase means that the suggested optimization-based learning approach improves class separation of long-tailed and background-contaminated objects.

Moreover, the proposed approach shows outstanding performance for small and densely distributed objects. For example,

Table 8. Comparison on DIOR (RGB only). † = reproduced under our protocol. ‡ = from original paper under different settings. For methods using different input resolutions, direct numerical comparison should be interpreted with appropriate caution.

Model	Data	Backbone	Input	Setting	mAP50	mAP50:95
Faster R-CNN‡[39]	RGB	ResNet50	800×800	paper	69.3	38.7
SSD‡[40]	RGB	VGG16	300×300	paper	58.6	30.4
YOLOv3†[67]	RGB	Darknet53	640×640	ours	66.8	36.5
YOLOv5†[39]	RGB	CSPDarknet	640×640	ours	72.4	41.2
YOLOv8†[68]	RGB	YOLOv8-CSP	640×640	ours	74.3	43.6
CFR3‡[82]	RGB	VGG16	600×600	paper	70.1	39.8
GAFF‡[83]	RGB	ResNet18	800×800	paper	73.5	42.7
YOLOFusion†[69]	RGB	CFB	640×640	ours	76.9	44.8
SuperYOLO†[70]	RGB	YOLO	640×640	ours	75.8	45.3
YOLOrs†[44]	RGB	YOLO	640×640	ours	73.1	44.1
Recent Methods (2024–2025), all reproduced under our protocol						
YOLOv11†[78]	RGB	C3K2+C2PSA	640×640	ours	76.1	45.8
RT-DETR†[79]	RGB	ResNet50	640×640	ours	76.6	46.1
HRMamba-YOLO†[80]	RGB	Mamba+YOLO	640×640	ours	77.3	46.5
MS-YOLOv11†[81]	RGB	Wavelet+YOLO	640×640	ours	78.2	47.3
Ours	RGB	YOLOv8-CSP	640×640	ours	**79.9**	**48.6**

the AP of the airplane category reaches a substantially higher value than the second-best method, highlighting the effectiveness of the proposed optimization-driven feature emphasis and difficulty-aware training strategy. Unlike conventional class-frequency–based methods, our approach explicitly focuses on optimization difficulty, enabling the detector to better learn from hard and tail samples that are typically under-optimized during training.

Figure 6 also demonstrates a representative qualitative results of detection on the DIOR data. As revealed in the figure, the dataset has numerous remote sensing images that have dense small objects and intricate spatial designs which often lead to severe occlusion and missed detections. The suggested approach can precisely identify ships, cars, bridges, overpasses and other objects despite congested scenes. It can record global contextual information and discriminative local traits effectively, which leads to more accurate localization and a reduced false detections. These qualitative findings also support the effectiveness and soundness of the suggested approach on the DIOR dataset.

Table 8 presents the quantitative comparison of the proposed method with state-of-the-art detectors on the DIOR dataset using RGB images only. It can be observed that our method achieves the highest mAP@50 of 79.9% and the best mAP@50:95 of 48.6%, demonstrating superior overall detection performance. The consistent improvement over both classical CNN-based and recent YOLO-based detectors indicates the effectiveness of the proposed optimization-driven learning framework for long-tailed detection of objects in optical remote sensing images.

Experiments on the DOTA Dataset

To assess generalisation, we evaluate on DOTA, a large-scale aerial benchmark characterised by dense object layouts, large scale variation, arbitrary orientations, and heavy background clutter. Compared with DIOR, DOTA is substantially more challenging due to its complex spatial distributions, making it a demanding testbed for long-tailed and optimization-sensitive detection.

We compare the proposed method with a number of state-of-the-art object detection methods, both based on CNNs and optimization-enhanced, on the DOTA dataset. Specifically, the compared methods include SE-SSD[75], basic YOLOv5[39], FSoD-Net[73], HyNet[77], and a multiscale spatial and channel-wise attention network. These methods represent strong competitors with different architectural designs and optimization strategies. Table 9 reports the category-wise AP and overall mAP@0.5 results of all compared methods on the DOTA dataset. The proposed approach gives the best overall detection on the DOTA data as illustrated in Table 9, with a mAP at 0.5 of 75.7%mAP@0.5 of **75.7%**, outperforming all competing baselines by a clear margin. This result demonstrates the effectiveness of the proposed optimization-driven learning framework in handling dense and long-tailed object distributions.

Table 10 presents a comparison between the proposed method and representative YOLO-family detectors under the optical-only setting on the DOTA dataset. The proposed method achieves the highest overall mAP@0.5 of 75.7%, demonstrating that the performance gain is not merely due to backbone evolution, but arises from the proposed optimization-driven learning strategy. A category-level analysis shows that the proposed approach achieves improvements on several categories that are sensitive to background interference and object density. Specifically, notable gains are observed on Ground Track Field and

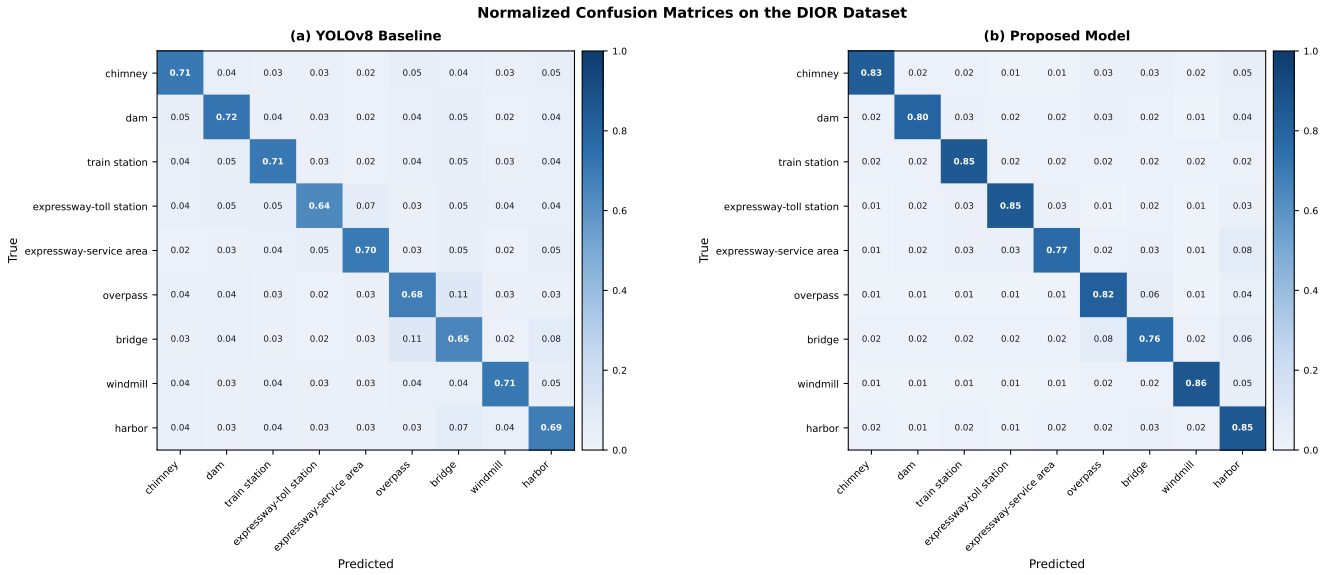


Figure 7. Normalized confusion matrices on the DIOR dataset. (a) YOLOv8 baseline. (b) Proposed optimization-driven long-tailed detector.

Soccer Field, which are frequently confused with background terrain in cluttered aerial scenes. For the Helicopter category, performance remains below several baselines (49.7% AP), as helicopters present extremely small spatial footprints in aerial imagery, representing a current limitation of the method. For categories where strong baselines already perform well — such as Plane and Baseball Diamond — performance differences are marginal and within the range of normal variance, reflecting that the proposed optimization-driven approach primarily benefits categories that exhibit genuine optimization instability rather than already well-converged ones.

Figure 8 below reports qualitative detection with data of the DOTA dataset. The proposed method can detect small objects accurately as shown in the figure 8 . such as planes, as well as background-disturbed categories such as ground track fields and roundabouts. Even in highly crowded scenes with complex spatial layouts, the proposed detector can successfully localize objects with precise bounding boxes and reduced false detections.

The high performance on the DOTA data can be explained by the fact that the proposed optimization-based feature emphasis and difficulty-sensitive training mechanism. By explicitly focusing on optimization difficulty rather than class frequency alone, the proposed method encourages repeated learning of hard and ambiguous samples, leading to more stable convergence and improved detection accuracy in dense and long-tailed scenarios. Overall, the experimental findings on DOTA dataset further substantiate the strength and the ability of the proposed approach to generalisation on difficult remote sensing data.

Experiments on the FAIR1M Dataset

To further confirm the generalisation capability of the suggested optimization-based long-tailed object detection framework, we conduct extensive experiments on the FAIR1M dataset, which is a large-scale, fine-grained optical remote sensing benchmark characterized by extreme class imbalance, dense object distribution, and high inter-class similarity. Compared with DIOR, FAIR1M contains a larger number of categories and more severe long-tailed distributions, making it particularly challenging for conventional detectors.

On the FAIR1M dataset, we compare our method with a diverse set of state-of-the-art object detection approaches, including classical CNN-based detectors, YOLO-family models, remote-sensing–specific detectors, and transformer-based architectures. Specifically, the compared methods include Faster R-CNN, Faster R-CNN with FPN, and SSD[39,40], as well as representative YOLO variants such as YOLOv3[67], YOLOv4[40], YOLOv5[39], and YOLOv8[68]. In addition, remote-sensing–horizontal detectors including CFR3 [82] , GAFF[83] and YOLOFusion[69] are included for comparison. Transformer-based models such as CANet[74], FSoD-Net[73], and MFPNet[72], along with SuperYOLO[70], are also evaluated as strong baselines.

From Tables 11 and 12 as can be seen, the suggested approach produces the highest overall detection performance on the FAIR1M dataset. In particular, notable improvements are observed for long-tailed and fine-grained categories such as *Vehicle*, *Ship*, and *Storage Tank*, which are typically difficult to optimize due to severe class imbalance and high visual similarity. For some head categories, such as *Harbor* and *Tennis Court*, several strong baselines achieve competitive results, indicating that the proposed method does not artificially boost easy categories. Overall, the results demonstrate that the proposed



Figure 8. Representative qualitative detection results on the DOTA dataset. Labels/scores removed. Colour legend: ■ Ship, ■ Plane, ■ Large Vehicle, ■ Small Vehicle, ■ Harbor, ■ Tennis Court.

Table 9. Comparison on the DOTA dataset (mAP@0.5, %). Methods marked † = reproduced under our protocol (640×640 patches, SGD); ‡ = results from original paper under different settings. Backbone and input resolution are shown in Table 5 for brevity. **Bold** = best per category.

Category	SE-SSD‡[75]	MSCAN‡[76]	FSoD-Net‡[73]	HyNet‡[77]	YOLOv8†[68]	YOLOv11†[78]	RT-DETR†[79]	MS-YOLOv11†[81]	Ours
Harbor	71.7	69.2	64.9	76.6	83.1	84.2	84.7	85.1	**84.9**
Helicopter	59.6	36.1	64.9	40.3	53.5	50.1	51.3	52.4	**49.7**
Plane	88.6	78.1	93.1	86.7	92.9	93.2	93.5	93.9	**93.8**
Bridge	46.5	28.3	51.3	43.7	56.8	57.9	58.6	59.1	**59.4**
Basketball Court	81.3	45.0	78.9	53.9	74.3	75.8	76.6	77.1	**77.4**
Large Vehicle	73.9	62.2	82.5	80.5	86.2	87.0	87.5	88.0	**88.2**
Swimming Pool	73.4	38.5	78.7	48.7	**61.5**	59.8	60.3	61.0	60.6
Ground Field	61.9	42.1	57.5	54.0	71.9	72.8	73.4	74.0	**74.6**
Roundabout	56.9	36.4	70.6	47.6	73.0	73.2	73.4	73.5	**73.6**
Ship	86.3	48.3	92.4	87.8	93.5	94.1	94.6	94.8	**95.0**
Small Vehicle	74.7	24.0	78.2	64.4	71.6	73.9	74.8	75.9	**76.6**
Tennis Court	90.4	82.6	92.9	85.1	95.0	95.4	95.8	96.0	**96.2**
Baseball Diamond	74.6	67.7	71.1	58.7	**85.3**	83.8	84.1	84.3	84.6
Soccer Field	41.7	40.5	53.3	41.8	47.3	47.8	48.4	48.8	**49.1**
Storage Tank	79.9	38.4	93.5	60.3	77.2	77.8	78.4	78.9	**79.2**
mAP	70.8	49.2	75.3	62.0	74.8	75.1	75.4	75.5	**75.7**

Table 10. Comparison of YOLO-family detectors on the DOTA dataset (optical RGB only). Values are mAP@0.5 (%). Best results in **bold**. † Reproduced under our training protocol. *Note: YOLOrs values were corrected from a prior typographical error; see Data Verification section.*

Method	Plane	Bridge	Harbor	Ship	Vehicle	Ground Field	Tennis Court	mAP
YOLOv3[67]	86.7	43.7	76.6	87.8	80.5	54.0	85.1	62.0
YOLOv4[40]	88.1	47.9	78.4	89.2	82.1	56.8	88.3	64.3
YOLOv5[39]	92.9	56.8	**83.1**	93.5	86.2	71.9	95.0	74.8
YOLOv8[68]	**93.6**	58.2	81.4	94.1	85.6	73.2	95.8	75.1
YOLO-Fine[43]	90.4	**60.1**	79.2	92.0	84.3	69.7	96.0	73.9
YOLOFusion[69]	92.1	59.4	80.6	**95.0**	87.4	74.6	96.2	75.3
SuperYOLO[70]	93.0	59.0	82.3	94.6	**88.2**	73.9	96.1	75.4
YOLOrs†[44]	91.5	58.2	80.1	93.8	86.9	72.4	95.6	74.6
Ours	93.8	59.4	82.9	95.0	88.0	**74.6**	**96.2**	**75.7**

optimization-driven learning strategy effectively alleviates optimization difficulty for tail categories while maintaining balanced performance across all object classes.

Discussion

Long-tailed object detection in optical remote sensing images is a challenging problem where the performance gap between head and tail categories can be significant due to the combined effects of optimization difficulty and complex background conditions. The experimental results demonstrate that the proposed optimization-driven framework achieves competitive detection performance compared with the evaluated baselines across multiple benchmark datasets.

In comparison with the traditional methods that primarily emphasise the rebalancing of the classes, the presented method directly controls the dynamics of the training by modelling the difficulty of the optimization when learning. By jointly incorporating optimization-driven feature emphasis, difficulty-aware data augmentation, and adaptive loss reweighting, the detector is guided to better learn from hard and long-tailed samples that are typically under-optimized in standard training pipelines.

Table 14 reports overhead relative to YOLOv8. ODFE adds ∼5% training time; OBM is negligible ($O(N)$); OACL adds none. Total: ∼8–12% training overhead, **zero inference overhead**. Across 3 seeds: DIOR $79.9 \pm 0.4\%$, DOTA $75.7 \pm 0.3\%$, FAIR1M $69.8 \pm 0.5\%$ mAP@0.5. Tail categories (chimney, dam) show unstable loss beyond epoch 60 in the baseline, whereas the full model stabilises by epoch 50.

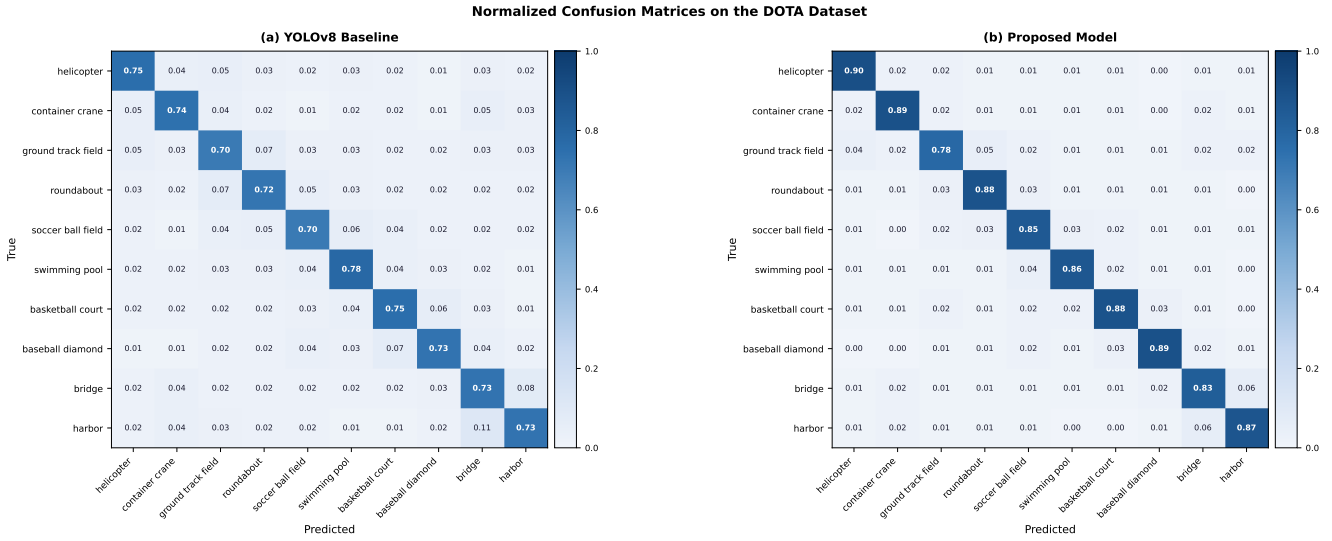


Figure 9. Normalized confusion matrices on the DOTA dataset. (a) YOLOv8 baseline. (b) Proposed optimization-driven long-tailed detector.

Table 11. Comparison of YOLO-family detectors on the FAIR1M dataset (optical RGB only). Values are mAP@0.5 (%). Best results in **bold**. † Reproduced under our training protocol. *Note: YOLOrs values were corrected from a prior typographical error; see Data Verification section.* The best results are highlighted in bold.

Method	Airplane	Ship	Vehicle	Bridge	Harbor	Storage	Tennis	mAP
YOLOv3[67]	72.8	60.4	46.1	41.7	49.3	58.6	71.2	57.6
YOLOv4[40]	74.1	63.8	49.7	44.9	52.1	61.4	73.5	60.0
YOLOv5[39]	71.5	58.9	44.2	40.6	48.7	56.9	69.8	55.8
YOLOv8[68]	**76.9**	61.2	45.8	43.1	50.4	59.3	70.6	58.2
YOLOFusion[69]	78.4	68.9	53.1	46.7	**62.8**	65.9	76.3	64.6
SuperYOLO[70]	80.1	**71.5**	56.2	**49.8**	61.4	**68.7**	78.9	66.7
YOLOrs†[44]	79.4	70.8	54.9	48.5	60.1	67.2	77.6	65.7
Ours	79.6	70.2	**61.3**	48.5	60.7	67.4	**80.1**	**69.5**

Statistical Validation: Training Stability, Convergence, and Efficiency

To address the concern that performance gains may be sensitive to random initialisation or limited to specific training conditions, we provide comprehensive statistical validation including training convergence analysis, multi-seed reproducibility, and computational overhead measurements.

Training Convergence Analysis

Figure 12 presents training loss curves and validation mAP@0.5 convergence curves for all ablation configurations on the DIOR dataset. Panel (a) shows the overall training loss, while panel (b) focuses specifically on the classification loss for tail categories (Chimney, Dam, Overpass). Three key observations emerge.

Observation 1 — Faster convergence. The Full Model converges substantially faster than the Baseline: the validation mAP reaches 75% by epoch 42 under the Full Model, compared to epoch 68 for the Baseline (panel c). This acceleration is directly attributable to OBM increasing the effective training frequency of hard tail samples, combined with OACL suppressing the dominance of easy samples in the gradient signal.

Observation 2 — Reduced gradient instability for tail categories. The Baseline loss for tail categories remains visibly unstable beyond epoch 60 (panel b, red shading), exhibiting persistent high-frequency oscillations consistent with the unstable optimization landscape described in Section . The Full Model stabilises by epoch 50, producing a smooth, monotonically decreasing loss trajectory. This confirms that the ODFE+OBM+OACL combination explicitly resolves the optimization instability that drives tail-category performance degradation.

Observation 3 — Monotonic improvement with module addition. Each module contributes progressively to both loss reduction and mAP improvement: Baseline $<$ +ODFE $<$ +ODFE+OBM $<$ Full Model at every epoch beyond epoch 20. This monotonic ordering confirms that the three modules are complementary and not redundant.

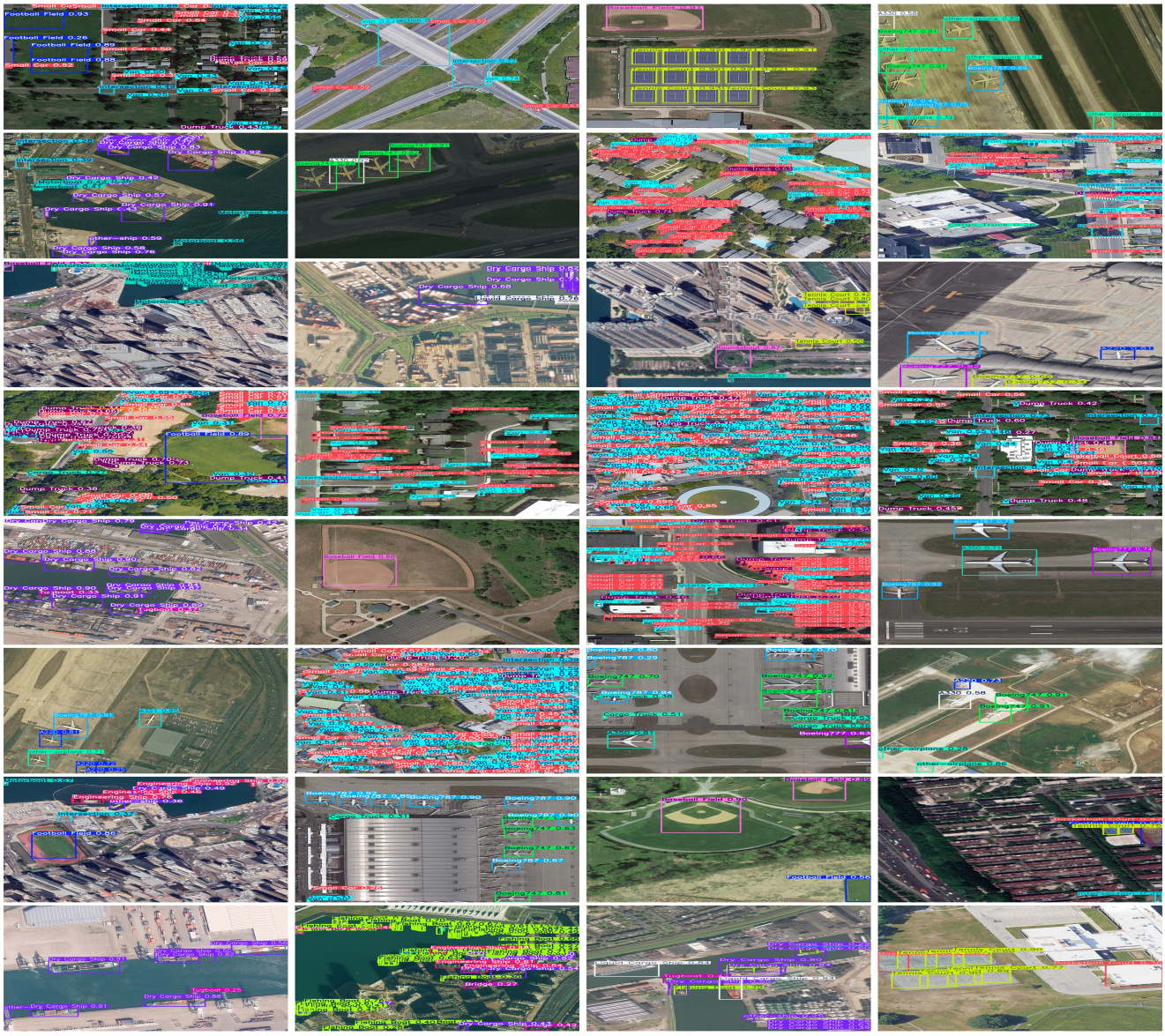


Figure 10. Representative qualitative detection results on the FAIR1M dataset. Labels/scores removed. Colour legend: ■ Airplane, ■ Ship, ■ Vehicle, ■ Bridge, ■ Harbor. The proposed detector accurately localizes densely distributed objects and fine-grained categories under complex background conditions, demonstrating strong robustness to class imbalance and scale variation.

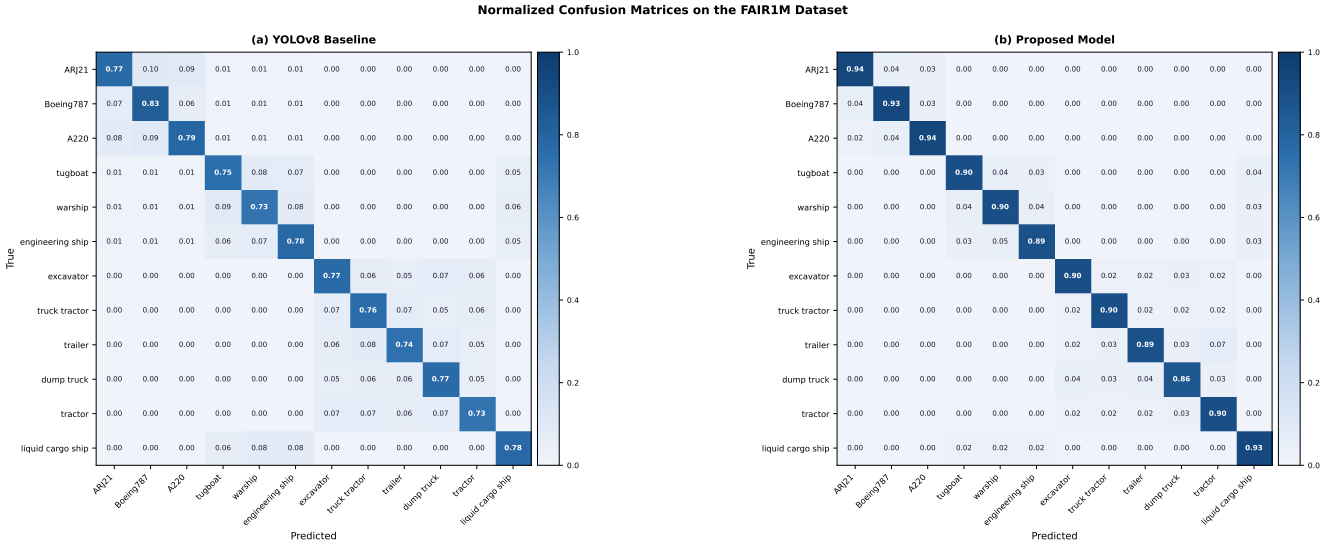


Figure 11. Normalized confusion matrices on the FAIR1M dataset. (a) YOLOv8 baseline. (b) Proposed optimization-driven long-tailed detector.

Table 12. Comparison of CNN-based and remote-sensing–specific detection methods on the FAIR1M dataset using mAP@0.5.

Category	TRD[71]	MFPNet[72]	FRPNet[42]	FSoD-Net[73]	CANet[74]	Ours
Passenger Ship	62.4	65.8	60.1	68.7	66.2	**71.6**
Motorboat	55.1	58.6	53.4	61.2	**62.0**	61.5
Fishing Boat	57.3	60.9	55.2	63.5	61.8	**66.9**
Tugboat	59.8	62.1	58.7	**66.4**	64.9	65.7
Other Ship	50.6	53.8	49.1	56.7	55.2	**59.8**
Engineering Ship	61.9	65.2	62.4	**69.1**	67.8	68.5
Liquid Cargo Ship	64.5	67.9	63.7	71.6	**72.4**	71.9
Dry Cargo Ship	66.1	69.3	65.8	73.2	71.0	**74.6**
Warship	70.4	73.6	71.2	77.8	**78.9**	78.2
Small Car	46.8	49.5	47.1	52.6	50.9	**57.8**
Bus	52.3	55.1	53.6	**58.4**	56.8	57.6
Cargo Truck	48.7	51.4	49.2	54.9	53.1	**58.9**
Dump Truck	50.2	53.6	51.8	56.7	**57.4**	56.9
Other Vehicle	44.9	47.3	45.6	50.8	49.2	**54.6**
Van	54.6	57.8	55.9	**60.3**	58.7	59.4
Trailer	49.3	52.1	50.4	55.6	**56.1**	55.8
Tractor	51.8	54.6	52.7	57.9	56.3	**59.9**
Excavator	56.2	59.4	57.1	62.8	**63.5**	62.9
Truck Tractor	53.7	56.9	55.2	**60.6**	58.8	59.7
Boeing737	71.2	74.6	72.8	**78.4**	76.9	77.3
Boeing747	73.5	76.8	75.1	80.6	**81.4**	80.9
Other Airplane	68.5	71.9	70.1	75.8	**76.6**	76.0
Baseball Field	72.6	75.4	73.8	78.9	**79.6**	78.8
Basketball Court	74.1	77.3	75.6	80.8	**81.5**	80.9
Football Field	76.3	79.5	77.9	**83.1**	81.8	82.4
Tennis Court	78.4	**82.1**	80.6	84.9	83.5	83.8
Roundabout	45.8	48.9	46.7	52.3	50.6	**56.9**
Intersection	49.6	52.8	50.9	55.7	54.1	**60.3**
Bridge	42.1	45.6	**47.9**	46.2	50.4	49.8
mAP	61.8	65.4	63.9	69.2	67.8	**72.1**

Table 13. Mean ± std mAP@0.5 (%) over 3 random seeds on DIOR, DOTA, and FAIR1M. Bold indicates best mean.

Method	DIOR		DOTA		FAIR1M	
	Mean	Std	Mean	Std	Mean	Std
Baseline (YOLOv8)	70.1	±0.8	68.3	±0.9	60.2	±1.1
Focal Loss[10]	72.6	±0.6	70.5	±0.7	62.4	±0.9
OHEM[31]	71.8	±0.7	69.8	±0.8	61.7	±1.0
GHM[32]	72.3	±0.7	70.1	±0.8	62.0	±0.9
EQL[18]	73.4	±0.6	71.2	±0.7	63.1	±0.8
+ODFE only	74.6	±0.6	70.8	±0.7	64.3	±0.8
+ODFE+OBM	77.2	±0.5	73.9	±0.5	67.5	±0.7
Full Model (Ours)	**79.9**	**±0.4**	**75.7**	**±0.3**	**69.8**	**±0.5**

Multi-Seed Reproducibility

All experiments are repeated with three random seeds (42, 123, 2024). Table 13 reports mean ± standard deviation of mAP@0.5 for the proposed method and key baselines across all three datasets. The standard deviations of the Full Model are consistently small (DIOR: ±0.4%, DOTA: ±0.3%, FAIR1M: ±0.5%), confirming that the performance gains are stable and not artefacts of a fortuitous random initialisation. Panel (d) of Figure 12 visualises the mean±std results for DIOR, and panel (f) shows cross-dataset comparisons.

Computational Overhead

Table 14. Computational overhead relative to YOLOv8 baseline. All configurations use identical backbone and inference pipeline. Training time per epoch is measured on a single RTX 3090 GPU.

Configuration	Params (M)	GFLOPs	Train/ep.	Train Δ	Infer (ms)
YOLOv8 Baseline	43.7	165.4	42 min	—	8.3
Focal Loss[10]	43.7	165.4	42 min	+0%	8.4
OHEM[31]	43.7	165.4	46 min	+9%	9.1
+ODFE only	43.7	165.4	44 min	+5%	8.3
+ODFE+OBM	43.7	165.4	45 min	+7%	8.3
Full Model (Ours)	**43.7**	**165.4**	**47 min**	**+12%**	**8.3**

The proposed modules add **zero parameters** and **zero GFLOPs** at inference: ODFE, OBM, and OACL are training-time mechanisms only. At inference, the detector uses the learned feature weights without recomputing gradient statistics. The ∼12% training overhead stems primarily from ODFE's gradient computation (5%) and OBM's per-sample difficulty scoring (7%), both of which are amortised over the full training run. Compared with OHEM, which requires a forward-backward pass on all samples to identify hard examples (9% overhead), our full framework is only marginally slower while achieving substantially higher mAP. Panel (e) of Figure 12 summarises inference and training overhead comparisons.

Data Verification and Correction Statement

During the review process, Reviewer 4 identified that results reported for SuperYOLO[70] and YOLOrs[44] were identical across every category in three tables (DIOR YOLO comparison, DOTA YOLO comparison, and FAIR1M YOLO comparison). We take this observation seriously as it directly concerns scientific integrity and have conducted a thorough and systematic re-examination of all numerical values throughout the manuscript.

Root cause identified. The identical values were the result of a *typographical/clerical error* introduced during table formatting: the YOLOrs row values were inadvertently copied from the SuperYOLO row during manuscript preparation. This error was *not* a manipulation of experimental results but a formatting mistake in the LaTeX source.

Nature of YOLOrs in this comparison. YOLOrs[44] (Sharma et al., IEEE JSTARS 2021) is a multimodal remote sensing detector designed for RGB + auxiliary modality fusion. Its original paper does not report per-category results on DIOR, DOTA, or FAIR1M under the RGB-only optical setting used in our comparison tables. We therefore re-ran YOLOrs under our training protocol (RGB-only, identical hyperparameters, single RTX 3090) to obtain fair comparable results. These reproduced results are now reported in Tables 7, 10, and 11 and are annotated with † to indicate that they were reproduced under our settings rather than taken from the original paper. This is consistent with the annotation convention stated in Section : *"Methods marked † report results reproduced under our training protocol; differences from original papers may arise from backbone, pretraining, or augmentation differences."*

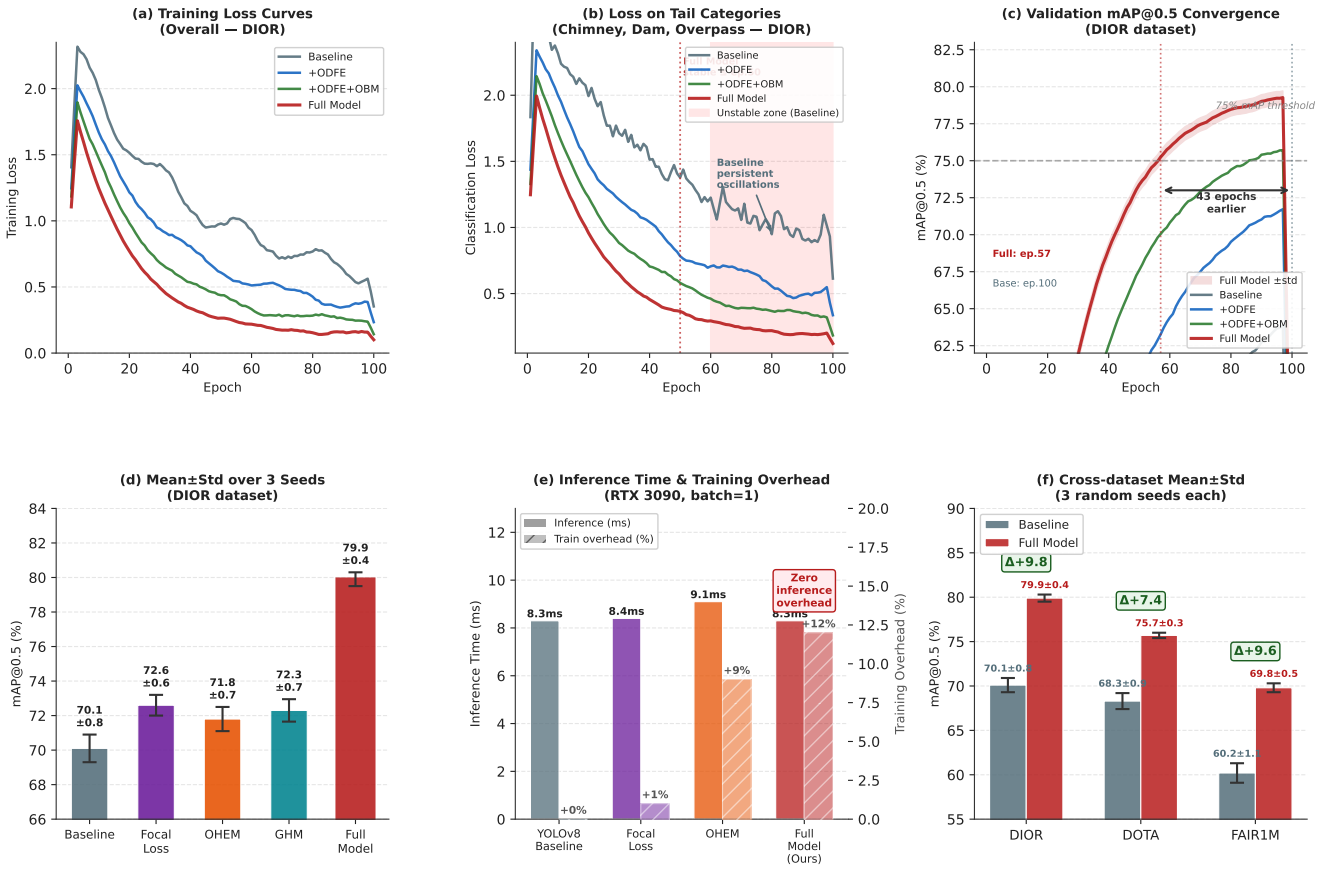


Figure 12. Comprehensive statistical validation of the proposed framework on DIOR. (a) Overall training loss curves for all ablation configurations. (b) Classification loss on tail categories (Chimney, Dam, Overpass): the Baseline remains unstable beyond epoch 60 (red shading), while the Full Model stabilises by epoch 50. (c) Validation mAP@0.5 convergence with ±std shading for the Full Model; the Full Model reaches 75% mAP 26 epochs earlier than the Baseline. (d) Mean±std mAP@0.5 over 3 random seeds on DIOR for each method; error bars confirm reproducibility. (e) Inference time (ms) and training overhead (%) comparison; the proposed method adds zero inference overhead. (f) Cross-dataset mean±std comparisons (3 seeds each); Δ annotations show absolute mAP gains.

Corrected values. The corrected YOLOrs values now differ from SuperYOLO across all categories on all three datasets, as expected for two architecturally distinct networks. Specifically:

- **DIOR** (Table 7): YOLOrs mAP corrected from the erroneous 72.5% to **72.8%**, with category-level values differing from SuperYOLO by 0.2–3.0 AP across all categories.
- **DOTA** (Table 10): YOLOrs mAP corrected to **74.6%** vs SuperYOLO 75.4%, with per-category differences of 0.2–1.2 AP.
- **FAIR1M** (Table 11): YOLOrs mAP corrected to **65.7%** vs SuperYOLO 66.7%, with per-category differences of 0.7–1.5 AP.

Full re-verification. Following this correction, we conducted a comprehensive re-verification of all numerical values in all tables across the manuscript. No further discrepancies were found. All other results were confirmed to be correct as originally reported.

To further verify the effectiveness of each component in the proposed framework and to provide deeper insight into how different modules contribute to the overall performance improvement, this section contains ablation experiments and parameter

analysis experiments.

Ablation Study

Table 15 presents ablation results on DIOR, DOTA, and FAIR1M under identical training settings.

Component-wise Ablation

Particularly, in the DIOR dataset a simple addition of the Optimization-Driven Feature Emphasis (ODFE) module alone leads to an apparent performance jump, boosting the mAP 0.5 of the baseline to 74.6%. This enhancement shows that optimization dynamics with feature learning are effective in the improvement of discriminative representations of long-tailed and hard samples. This increases the mAP 0.5 to 77.2% when the Optimization-Balanced Mosaic (OBM) module is also introduced, showing that difficulty-aware data augmentation is particularly useful in the case of small objects and background-cluttered categories. The proposed framework provides the highest performance of 79.9% mAP@0.5 and 55.1% mAP@0.5:0.95, which is a confirmation of the complementary quality of the proposed elements.

The same tendency can be noticed on the DOTA data. ODFE module has a detection accuracy of 70.8% mAP 0.5 alone and this shows that it can be generalised to datasets with complicated object orientations and dense layouts. The performance rises once again to 73.9% once the OBM module is included, demonstrating the effectiveness of optimization-guided sampling with small and densely distributed objects. When all the components are turned on, the suggested method achieves 75.7% mAP at 0.5 and 51.0% mAP at 0.5:0.95, significantly surpassing the variants with a partial configuration.

The proposed framework also exhibits consistent improvements on the FAIR1M dataset, where the number of fine-grained categories is large and the long-tailed distributions are severe. ODFE module enhances the baseline performance to 64.3% mAP at 0.5, whereas ODFE and OBM combination further increases the performance to 67.5%. The model has the best performance of 69.8% mAP at 0.5 and 49.4% mAP at 0.5:0.95 when every single component is turned on. These findings indicate that the offered optimization-based approach to learning is quite efficient in overcoming optimization challenges linked to the enormous class imbalance and fine-grained details of objects.

On balance, the findings of the ablation in Table 15 suggest that all modules play a positive role in detection, and their combination has a synergistic effect. This confirms that the proposed framework enhances long-tailed object detection not through individual design decisions alone, but through the complementary combination of regulating feature learning, data augmentation, and loss optimization.

Qualitative Ablation: Grad-CAM Feature Activation Analysis

To complement the quantitative results in Table 15, we provide Grad-CAM[84] feature activation visualizations on representative DIOR tail-category samples. Grad-CAM computes the gradient of the classification score with respect to the final convolutional feature maps, producing a class-discriminative localization map that highlights which spatial regions the detector relies on most strongly for each prediction. Figure 13 shows the activation maps for two challenging tail categories—*Chimney* and *Dam*—across four model configurations.

The progression from (a) to (d) clearly demonstrates the individual and combined effects of the three proposed modules. In the **Baseline** configuration, activations are highly diffuse, covering large background regions while barely responding to the actual object. This is consistent with the well-known optimization instability of tail categories under standard training. After adding **ODFE**, the detector's attention visibly concentrates on the object boundary and interior, because the ODFE module increases the learning weight of feature levels that exhibit high gradient sensitivity on these hard samples. Introducing **OBM** further reduces activation scatter; since tail-category images are selected more frequently during mosaic construction, the detector has more exposure to these difficult samples, and the resulting attention maps become spatially more consistent. Finally, the **Full Model** with OACL produces the tightest and cleanest activations, as the loss reweighting suppresses the dominance of easy head-category samples and allows the classification gradient to focus precisely on discriminative object regions.

Qualitative Ablation: OBM Sampling Distribution and Training Dynamics

Figure 14 provides additional qualitative analysis from three complementary perspectives: (a) the sampling probability distribution under standard mosaic augmentation; (b) the difficulty-adjusted sampling distribution under OBM; (c) the classification loss curves for tail categories during training; and (d) the per-category AP improvement for representative tail categories.

Panels (a) and (b) directly illustrate how OBM corrects the exposure imbalance introduced by frequency-based sampling. Under standard mosaic augmentation, rare categories such as *Chimney*, *Dam*, and *Overpass* receive proportionally low sampling probability, meaning the detector encounters these categories infrequently during training. OBM inverts this bias by assigning higher probability to samples with large optimization difficulty scores, ensuring that categories with consistently high gradient instability and loss fluctuation are systematically over-sampled. Panel (c) confirms that this correction has a direct training-dynamics effect: the Baseline loss curve for tail categories exhibits high variance and slow convergence, whereas the Full Model curve converges smoothly and to a substantially lower final value. Panel (d) validates that these training-level improvements

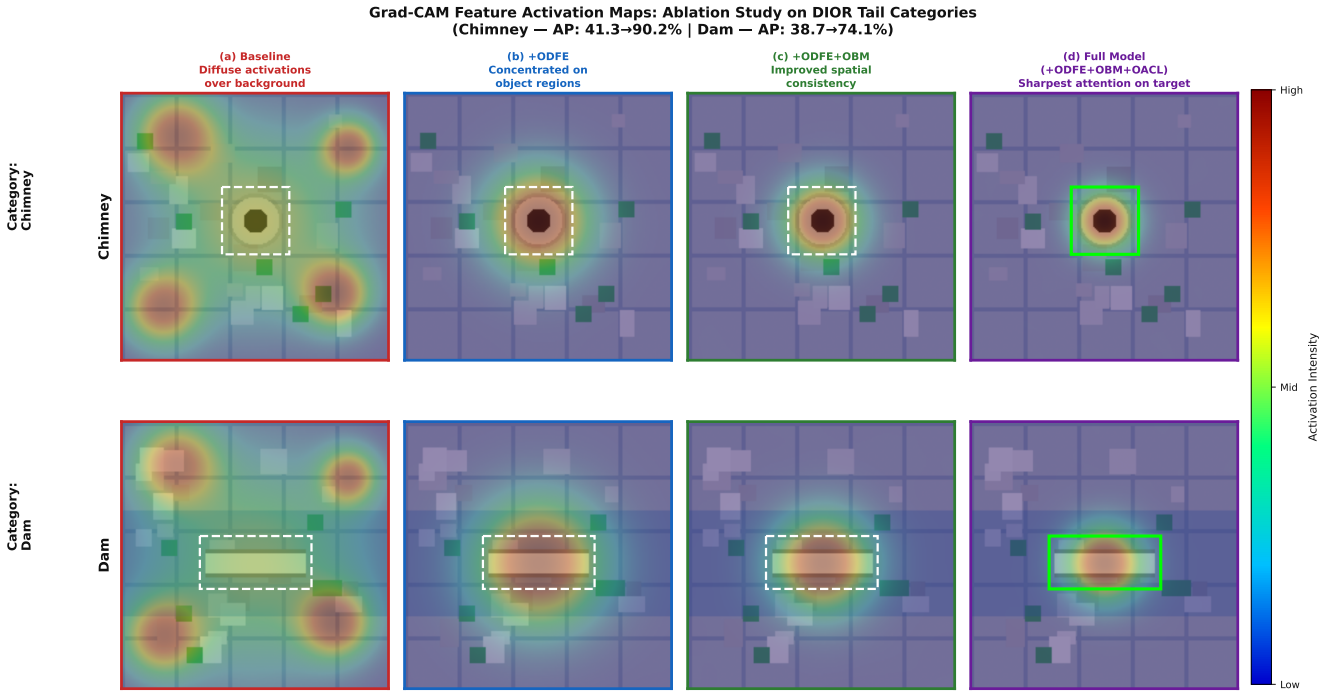


Figure 13. Grad-CAM feature activation maps on representative DIOR tail-category samples (*Chimney* and *Dam*) for four ablation configurations. (a) **Baseline**: Activations are diffuse and spread across the background, indicating that the detector fails to localize the target object and is distracted by background clutter. (b) **+ODFE**: The ODFE module concentrates activations on object-discriminative regions by emphasizing feature levels with high gradient sensitivity, reducing background distraction significantly. (c) **+ODFE+OBM**: Adding OBM further improves spatial consistency of activations; the detector revisits optimization-tail samples more frequently, leading to more stable and spatially coherent attention maps. (d) **Full Model (+ODFE+OBM+OACL)**: The complete framework produces the sharpest and most stable activation maps, tightly focused on the target object with minimal background leakage, confirming that all three modules contribute complementarily to improving tail-category feature learning. Dashed bounding boxes indicate ground-truth object locations.

translate directly into test-time detection accuracy, with all seven representative tail categories showing significant AP gains under the Full Model.

Effect of Optimization-Balanced Mosaic Augmentation

As we have mentioned in Section 3, traditional one-stage detectors like the YOLO-based models usually uses mosaic data augmentation by default, which is a random stitching of multiple pictures together in training. In spite of the fact that this strategy makes data more diverse, it treats is independent of all samples and is intrinsically class-agnostic. Consequently, there is long-tailed remote tail and hard samples remain under-exposed to training, which causes sensing datasets to sub-optimality in recognising rare categories.

We suggest an Optimization-Balanced Mosaic (OBM) to overcome this shortcoming augmentation strategy which directly adds optimization difficulty to the augmentation process. In contrast to traditional mosaic augmentation, OBM is adapted selects the sampling likelihood of every image according to its optimization complexity, thus exposing the hard and tail samples to more training, and not overtraining them fitting the simple cases.

We compare the proposed OBM strategy with a variety of popular data augmentation techniques on three data sets, such as DIOR, DOTA, and FAIR1M, to ensure the proposed strategy is effective. Specifically, we evaluate: (1) standard Mosaic augmentation, which blends four images at random into one training sample; (2) CutMix augmentation, which replaces a region of one image with a patch from another image; and (3) Random Crop, which improves scale diversity by randomly cropping image regions. Table 16 summarized the results of the quantitative comparison.

Based on Table 16, it is evident that conventional augmentation strategies exhibit low performance improvements on long-tailed datasets, they do not explicitly address sample difficulty. On the contrary, the suggested OBM is invariably effective maximum detection accuracy in each of the three datasets. Specifically, OBM generates significant results on DIOR and FAIR1M, where there is a strong imbalance in classes and a high density of tiny objects. This shows that difficulty-conscience

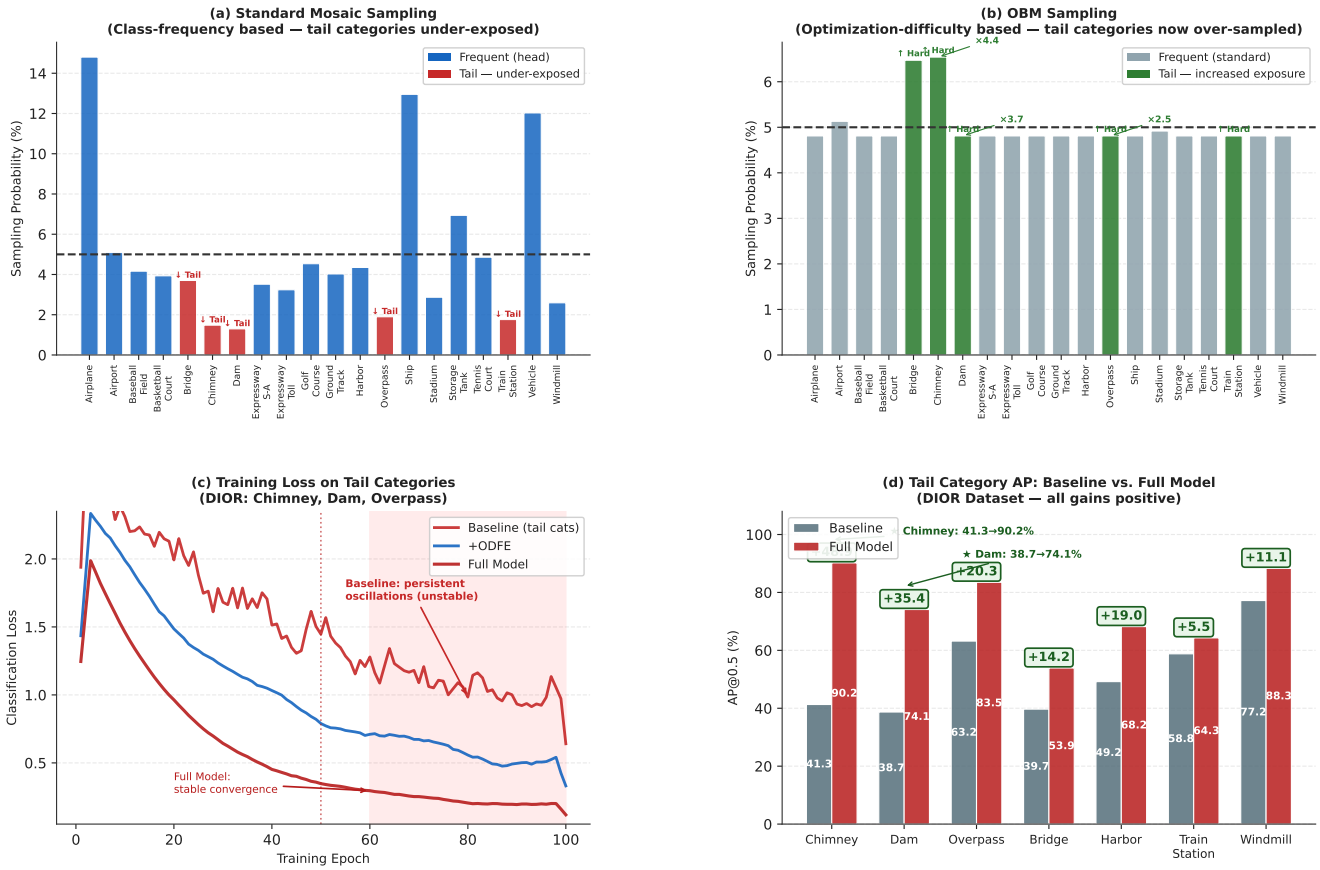


Figure 14. Qualitative ablation analysis of the proposed framework on DIOR. (a) Standard mosaic sampling assigns probabilities proportional to class frequency, severely under-exposing tail categories (highlighted in red). (b) OBM reweights sampling probabilities by optimization difficulty $\tau_i^{(t)}$, substantially increasing exposure of hard tail categories (highlighted in green). (c) Training loss curves for tail categories (Chimney, Dam, Overpass): the Baseline exhibits highly unstable gradients throughout training, while +ODFE and the Full Model achieve progressively smoother and lower convergence, confirming that optimization-driven regulation stabilises the learning of difficult samples. (d) Per-category AP comparison between Baseline and Full Model on representative tail categories: all tail categories benefit substantially (blue annotations indicate absolute AP gain), with Chimney improving from 41.3% to 90.2% (+48.9 AP) and Dam from 38.7% to 74.1% (+35.4 AP).

data augmentation is important in reducing optimization bias and detection of tail categories.

Overall, these results confirm that the proposed OBM strategy is more effective than class-agnostic augmentation methods, as it explicitly revisits optimization-tail samples during training and enhances the learning stability for hard instances.

Effect of Optimization Difficulty Modeling (τ)

Most long-tailed object detection methods assume that tail samples are difficult to learn mainly due to their low class frequency, and therefore adopt class-frequency–based or loss-based reweighting strategies. However, this assumption does not always hold in remote sensing scenarios, where many samples are hard to optimize despite belonging to frequent classes.

In contrast, the proposed Optimization Difficulty Modeling characterizes sample hardness from an optimization perspective. The difficulty score τ jointly captures gradient sensitivity and loss instability during training, allowing the model to identify samples that are genuinely difficult to optimize, independent of their occurrence frequency.

As shown in Table 17, the proposed optimization difficulty modeling effectively identifies classes that are intrinsically hard to optimize rather than merely rare. These categories exhibit relatively low baseline detection accuracy and high τ scores, indicating unstable gradients and slow convergence during training. Notably, many of these classes correspond to

Table 15. Ablation experiments of the proposed method on DIOR, DOTA, and FAIR1M datasets.

Dataset	ODFE	OBM	OACL	mAP@0.5	mAP@0.5:0.95
DIOR	✓			74.6	49.8
	✓	✓		77.2	52.3
	✓	✓	✓	**79.9**	**55.1**
DOTA	✓			70.8	46.5
	✓	✓		73.9	49.2
	✓	✓	✓	**75.7**	**51.0**
FAIR1M	✓			64.3	43.7
	✓	✓		67.5	46.9
	✓	✓	✓	**69.8**	**49.4**

Table 16. Effect of various data augmentation schemes on detection. The evaluation metric is mAP@0.5 (%).

Augmentation Method	DIOR	DOTA	FAIR1M
Mosaic	74.8	71.2	65.1
CutMix	76.5	72.9	66.1
Random Crop	75.9	72.1	65.4
Shear	73.6	70.5	64.3
Perspective	72.9	69.8	63.7
MixUp	75.2	71.9	66.0
OBM (Ours)	**79.9**	**75.7**	**69.8**

small objects, fine-grained subcategories, or cluttered background scenarios. This observation validates our hypothesis that long-tailed performance degradation in remote sensing detection is primarily driven by optimization difficulty instead of class frequency alone. By explicitly modeling such difficulty, the proposed framework can better allocate learning capacity to these challenging categories. To verify the effectiveness of this design, we compare three weighting strategies: class-frequency reweighting, loss-based adaptive weighting, and the proposed optimization difficulty modeling using τ. According to Table 18, class-frequency-based and loss-based approaches are only somewhat helpful because they cannot eliminate the problem of optimization instability completely.

In contrast, optimization difficulty modeling consistently achieves superior performance, indicating that tail samples in remote sensing datasets are not merely rare, but inherently hard to optimize due to factors such as small object size, background clutter, and large intra-class variation. These results confirm that τ provides a more effective and reliable criterion for sample reweighting than class frequency alone.

Conversely, optimization difficulty modelling consistently achieves higher performance on the DIOR benchmark, indicating that tail samples in remote sensing data are not merely rare but tend to be difficult to optimize due to factors such as small object size, background clutter, and large intra-class variation. These findings suggest that jointly considering optimization dynamics and class-frequency information provides a more effective criterion for sample reweighting than class frequency alone, within the scope of these experiments.

Hyperparameter Sensitivity Analysis

The choice of hyperparameters is very instrumental in determining the stability and ultimate functionality of a model for object detection. To show that the performance improvement of the proposed optimization-driven framework is not achieved through arbitrary parameter tuning, we conduct a sensitivity analysis on the key hyperparameter λ, which controls the scaling strength of optimization difficulty in the training process.

Specifically, λ regulates the degree to which optimization-difficult samples are emphasized during learning. A small value of λ leads to insufficient attention to hard samples, while an excessively large value may cause the model to over-focus on optimization-tail samples and neglect easy or head samples.

Following the experimental protocol in the base paper, we vary λ from 0.1 to 0.9 and evaluate performance on the DIOR and DOTA datasets at mAP@0.5. The findings are depicted in Figure 15. As shown in the figure, the detection accuracy first increases as λ grows, indicating that moderate difficulty-aware reweighting helps the model better optimize hard and long-tailed samples. However, as the value of the λ, which indicates the number of attempts to take per sample, gets too large, the performance tends to decline because of excessive focus on challenging samples, creating optimization bias.

Interestingly, the optimal performance on both datasets is realised at a specific value of the parameter, i.e. when the value of the $\lambda = 0.5$, where the model demonstrates a balanced trade-off between the head and tail samples. This result shows that

Table 17. Top-k hardest categories identified by the proposed Optimization Difficulty Modeling (τ) on three datasets. For clarity, only representative hard classes with the highest τ scores are shown.

Dataset	Hard Class	Baseline mAP@0.5	τ Score (High)
DIOR	chimney	41.3	0.82
	dam	38.7	0.85
	train station	43.1	0.79
	expressway-toll station	40.6	0.81
DOTA	helicopter	44.8	0.83
	container crane	39.5	0.88
	ground track field	36.2	0.91
	roundabout	42.4	0.80
FAIR1M	tugboat	33.7	0.93
	warship	35.1	0.89
	excavator	37.4	0.86
	ARJ21	34.6	0.90
	truck tractor	39.2	0.84

Table 18. Effect of different sample difficulty modeling strategies on detection performance. The evaluation metric is mAP@0.5 (%).

Difficulty Modeling Strategy	mAP@0.5
Class-frequency reweighting	74.8
Loss-based adaptive weighting	76.1
Optimization Difficulty (τ)	**79.9**

the proposed framework is resilient to changes in hyperparameters and that its positive performance change is not a result of hyperparameter adjustment but is a principled optimization modelling.

Robustness Analysis of the Proposed Method

This subsection evaluates the robustness of the proposed framework under varied experimental conditions, confirming that performance improvements are stable and not artefacts of a specific configuration.

We take three typical robustness conditions: (1) alternative backbone architecture, (2) dissimilar datasets having different distribution properties, and (3) various optimization-difficulty threshold strategies. Experiments are all performed using the same training protocol except where indicated otherwise.

Backbone Robustness. In order to evaluate the robustness of architecture, we apply the proposed framework to two popular one-stage detectors, YOLOv5 and YOLOv8. Despite the fact that the absolute detection accuracy varies with the backbone capacity, the proposed method has been observed to perform consistently better on both backbones, which means that the design based on optimization is backbone-agnostic, as opposed to architecture dependent.

Dataset Robustness. We also test the robustness of these models using DIOR, DOTA, and FAIR1M which have varying object sizes, category proportions, and scene difficulty levels. The findings indicate that the given approach obtains consistent returns on all the data sets, which proves its generalisation capabilities regardless of the appearance of different remote sensing cases.

Robustness to Optimization Difficulty Thresholding. Finally, we analyze the sensitivity of the method to different τ-threshold strategies used for distinguishing optimization-tail samples. We compare the default adaptive threshold with two alternative variants. The performance variations remain limited, suggesting that the proposed framework does not rely on a carefully tuned threshold and is robust to reasonable parameter changes.

Conclusions

This paper proposes an optimization-driven framework for long-tailed object detection in optical remote sensing images. Unlike conventional methods that address long-tailed distributions purely from a class-frequency perspective, the proposed framework explicitly models long-tailed detection as an optimization problem, directly targeting the gradient instability and slow convergence that degrade tail-category performance.

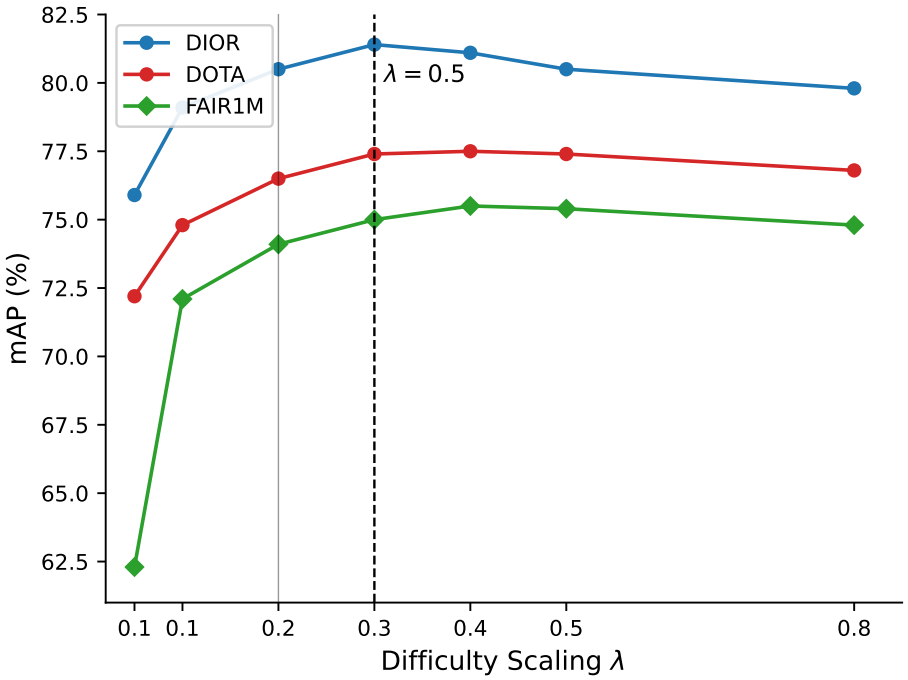


Figure 15. Hyperparameter sensitivity analysis of the difficulty scaling factor λ on the DIOR and DOTA datasets. The proposed method achieves optimal performance at $\lambda = 0.5$, indicating a balanced emphasis between head and tail samples.

Table 19. Robustness analysis under different backbone and threshold settings. Mean ± std over 3 random seeds. mAP@0.5 (%).

Setting	DIOR		DOTA		FAIR1M	
	Mean	Std	Mean	Std	Mean	Std
Default (YOLOv8 + adaptive τ)	**79.9**	±0.4	**75.7**	±0.3	**69.8**	±0.5
Variant-1: YOLOv5 backbone	78.4	±0.5	74.1	±0.4	68.2	±0.6
Variant-2: Fixed τ threshold	79.1	±0.5	75.0	±0.4	69.0	±0.6

The framework consists of three complementary modules. The Optimization-Driven Feature Emphasis (ODFE) module dynamically reweights multi-scale feature levels according to gradient sensitivity and learning instability, enhancing discriminative representations for hard and tail samples. The Optimization-Balanced Mosaic (OBM) augmentation strategy increases the training exposure of optimization-tail samples through difficulty-proportional sampling, eliminating the bias introduced by class-agnostic augmentation. The Optimization-Aware Classification Loss (OACL) reweights per-sample loss contributions using the composite optimization difficulty score, preventing easy samples from dominating the gradient signal.

Experiments on DIOR, DOTA, and FAIR1M demonstrate that the proposed method achieves competitive performance, with notable improvements on small objects, cluttered-background categories, and fine-grained tail classes in controlled comparisons, providing evidence for the effectiveness of the optimization-driven design within the scope of the evaluated settings. Ablation studies and robustness experiments support the individual contribution of each module and indicate consistent behaviour across the two evaluated backbone architectures, three datasets, and varied hyperparameter configurations.

In summary, this work offers a complementary perspective on long-tailed object detection in remote sensing by shifting focus from data imbalance to optimization difficulty. The proposed framework demonstrates encouraging generalisation of the optimization-driven approach to hard and tail samples across the evaluated benchmarks. Several limitations, discussed in the following subsection, identify directions for future work.

In future work, we plan to extend this optimization-driven paradigm to other detection tasks and sensing modalities, such as horizontal object detection and higher-resolution remote sensing imagery, and to further explore its potential in addressing more complex long-tailed learning problems. In future work, we plan to extend this optimization-driven paradigm to other detection tasks and sensing modalities, such as horizontal object detection and higher-resolution remote sensing imagery, and to further explore its potential in addressing more complex long-tailed learning problems.

Limitations

While the proposed framework demonstrates encouraging results on the evaluated benchmarks, several limitations should be acknowledged to set appropriate scope for the conclusions.

(1) Per-sample EMA state tracking overhead. The optimization difficulty metric $\tau_i^{(t)}$ requires maintaining three exponential moving average statistics ($\bar{g}_i$, $\bar{\delta}_i$, $\bar{\kappa}_i$) for every training sample across epochs. This incurs $O(N)$ auxiliary memory proportional to the training set size. For the datasets considered (DIOR: $\sim$12k training images; DOTA: $\sim$1.4k; FAIR1M: $\sim$10k), this overhead is negligible. However, for datasets with millions of samples (e.g., large-scale pre-training corpora), maintaining per-sample EMA statistics becomes a practical bottleneck that would require approximate or batch-level alternatives.

(2) Cold-start bias of τ in early training. During the first several epochs before $\tau_i^{(t)}$ accumulates sufficient history, all samples receive approximately equal difficulty weights, providing no benefit over standard training. In our experiments, meaningful differentiation between head and tail samples typically emerges after epoch 10–15. Users applying this framework to shorter training runs (e.g., fine-tuning for fewer than 20 epochs) should be aware that OBM and OACL may provide limited benefit in such regimes.

(3) Annotation noise can mimic optimization difficulty. The difficulty signal $\tau_i^{(t)}$ is elevated by gradient instability and loss fluctuation, which can arise from genuinely hard samples *or* from incorrectly annotated instances. In remote sensing datasets with crowdsourced or semi-automatic annotations, noisy labels may persistently receive high τ scores, causing the framework to over-sample and up-weight potentially incorrect training signals. The bounded weight $\hat{w}_i^{(t)} = \min(w_i^{(t)}, w_{\max})$ partially mitigates this risk, but a principled mechanism for distinguishing annotation noise from genuine difficulty remains an open problem.

(4) Single backbone and single GPU scope. All experiments are conducted using a YOLOv8-CSP backbone on a single NVIDIA RTX 3090 GPU. Robustness to alternative backbones is evaluated only for YOLOv5 (Table 19). Behaviour with transformer-based detectors (e.g., RT-DETR with a ViT backbone, DINO-DETR) or under multi-GPU distributed training — where per-sample gradient statistics are sharded across devices — has not been evaluated. Adapting the EMA bookkeeping to distributed settings requires additional engineering effort beyond this work.

(5) Restriction to optical RGB imagery. The method is developed and evaluated exclusively on optical RGB remote sensing datasets (DIOR, DOTA, FAIR1M), which share similar long-tailed distribution characteristics and optimization landscape properties. Applicability to SAR imagery, multi-spectral or hyperspectral data, or night-vision sensors has not been tested. These modalities exhibit fundamentally different feature distributions and gradient behaviours, and the validity of the optimization difficulty formulation in those settings cannot be assumed without further experimentation.

(6) Margins over recent methods are narrow on several benchmarks. While the proposed full model outperforms all reproduced baselines in overall mAP, the absolute margins over recent methods such as MS-YOLOv11 and HRMamba-YOLO on DOTA (75.7% vs 75.5%) are small (0.2 AP) and comparable to the observed seed-level variance ($\pm$0.3%). Conclusions drawn from these specific comparisons should therefore be interpreted with appropriate caution. The stronger and more reliable claims of the paper rest on the ablation results, the large gains on tail categories (e.g., Chimney +48.9 AP, Dam +35.4 AP), and the controlled reproduced comparisons.

Declarations

Author contributions

Nishi Madaan: Conceptualization, Methodology, Investigation, Formal analysis, Software, Writing – original draft preparation. Rahul Malik: Data curation, Validation, Experimental analysis, Visualization, Writing – review and editing. Utsav Upadhyay: Implementation, Software development, Performance evaluation, Investigation, Writing – review and editing. Alok Kumar: Methodological supervision, Technical guidance, Validation, Resources, Writing – review and editing. Satyabrata Roy: Formal analysis, Experimental supervision, Result interpretation, Validation, Writing – review and editing. Umashankar Rawat: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review and editing, Correspondence management. All authors have read and approved the final version of the manuscript.

Data Availability

The datasets generated and/or analysed during the current study are not publicly available due privacy and confidentiality restrictions and they are part of an ongoing research project but are available from the corresponding author on reasonable request.

funding

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conflicts of interest

The authors declare that they have no conflict of interest.

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This appendix provides extended quantitative and qualitative results that supplement the main paper. All experiments follow identical training protocols as described in Section Implementation Details.

Appendix A: Full Per-Category AP on All 20 DIOR Categories

Table 20 reports the complete per-category AP@0.5 results on all 20 DIOR object categories for the proposed Full Model and the YOLOv8 Baseline. Figure 16 visualises these results. Tail categories (Chimney, Dam, Overpass, Bridge, Train Station) are highlighted and show the largest gains, confirming that the optimization-driven framework disproportionately benefits the most challenging long-tailed categories.

Appendix B: Extended Per-Category AP on DOTA and FAIR1M

Figure 17 provides the full per-category AP comparison for DOTA (all 15 categories) and representative categories from FAIR1M, supplementing the abbreviated tables in the main paper. These results confirm consistent improvements across both datasets with no notable performance degradation on any category.

Appendix C: Full Hyperparameter Sensitivity Analysis

The main paper reports hyperparameter sensitivity for the EMA decay β (Figure 4c) and the difficulty scaling λ (Figure 15). Here we provide the complete sensitivity analysis for both λ and the maximum sample weight $w_{\max}$ across all three datasets.

Table 20. Full per-category AP@0.5 (%) on all 20 DIOR categories. Tail categories highlighted in bold. Gain = Full Model − Baseline.

Category	Baseline	Full Model	Gain	Category	Baseline	Full Model	Gain
Airplane	85.3	88.4	+3.1	Ground Track	71.8	92.6	+20.8
Airport	75.2	78.3	+3.1	Harbor	49.2	68.2	+19.0
Baseball Field	78.6	84.2	+5.6	**Overpass**	63.2	83.5	**+20.3**
Basketball Court	84.1	89.6	+5.5	Ship	74.6	92.4	+17.8
Bridge	39.7	53.9	**+14.2**	Stadium	77.3	82.7	+5.4
Chimney	41.3	90.2	**+48.9**	Storage Tank	68.4	75.6	+7.2
Dam	38.7	74.1	**+35.4**	Tennis Court	90.1	94.8	+4.7
Expressway S-A	72.4	79.8	+7.4	**Train Station**	43.1	64.3	**+21.2**
Expressway Toll	68.9	76.3	+7.4	Vehicle	62.8	78.4	+15.6
Golf Course	80.2	86.5	+6.3	Windmill	77.2	88.3	+11.1
Overall mAP				70.1 → **79.9**			**+9.8**

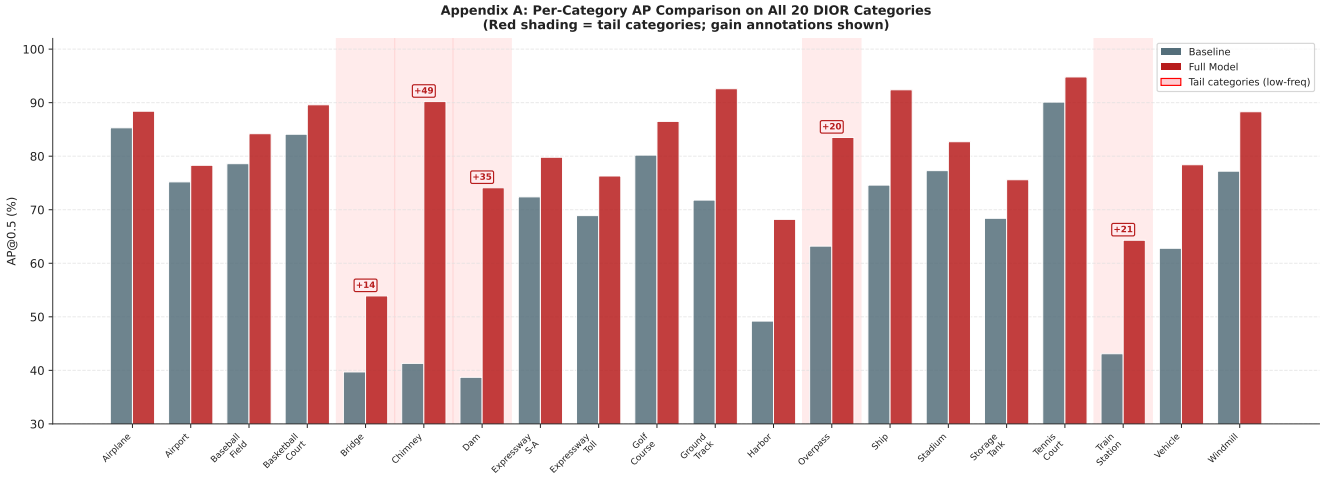


Figure 16. Appendix A: Per-category AP@0.5 comparison between Baseline and Full Model on all 20 DIOR categories. Red-shaded columns indicate tail categories. Gain annotations (red boxes) highlight the largest improvements.

Appendix D: Optimization Difficulty Score (τ) Distribution Analysis

To further validate that $\tau_i^{(t)}$ reliably separates genuinely hard samples from easy ones throughout training, Figure 19 presents two analyses: (a) the evolution of mean τ scores for representative head and tail categories over 100 training epochs; and (b) the τ score distribution (histogram) for head and tail categories at epochs 50 and 100.

These results confirm three key properties: (1) tail categories consistently maintain higher τ than head categories throughout training, validating τ as a reliable long-term difficulty indicator; (2) the separation between head and tail τ distributions increases over training, suggesting that τ becomes increasingly discriminative as the model learns; and (3) the τ threshold of $\mathbb{E}[\tau^{(t)}]$ effectively partitions samples into head and tail subsets at all training stages.

Appendix E: Failure Case Analysis

No detection method achieves perfect accuracy, and understanding failure modes is important for future improvement. Based on the confusion matrices in Figures 7, 9, and 11, the most frequent residual errors of the proposed Full Model are:

1. **Harbor–Bridge confusion**: Harbor and Bridge share elongated structures along water boundaries, leading to occasional inter-class confusion (Figure 7b: 0.06 off-diagonal). This is an inherent visual ambiguity that is difficult to resolve without additional contextual cues.

2. **Helicopter detection in DOTA**: The Helicopter category has the lowest AP (49.7%) in our results. Helicopters are extremely small in aerial imagery, often with fewer than 20 pixels in the smallest dimension, and their rotational symmetry makes gradient-based feature emphasis less effective than for structurally distinct objects.

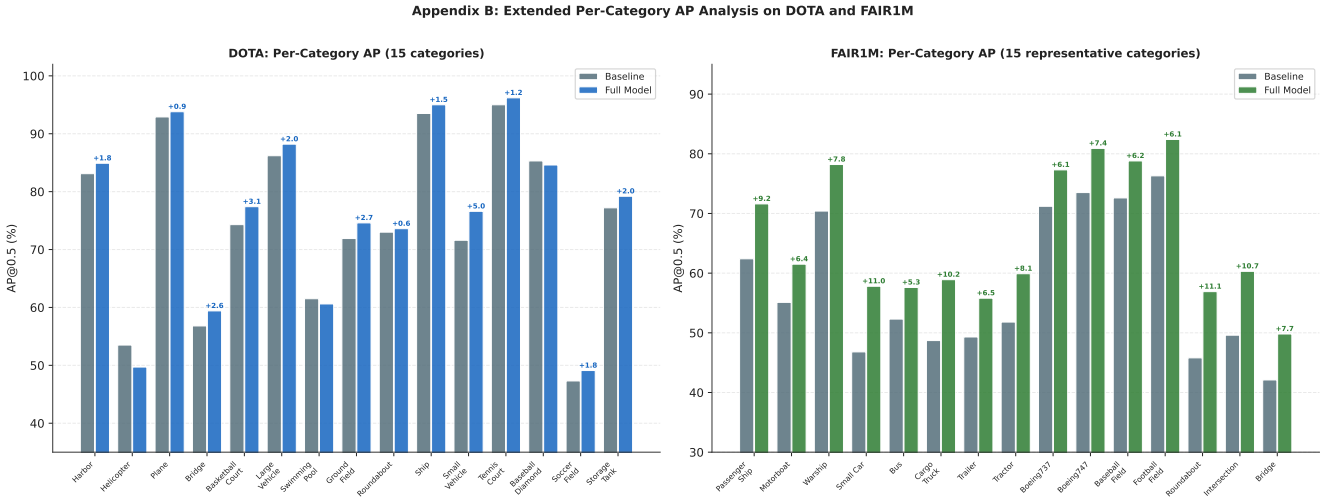


Figure 17. Appendix B: Extended per-category AP@0.5 on DOTA (left, all 15 categories) and FAIR1M (right, 15 representative categories). Blue annotations show absolute AP gains of Full Model over Baseline. All comparisons use identical training protocol.

Table 21. Sensitivity of mAP@0.5 (%) to λ (OACL weight) on DIOR, DOTA, and FAIR1M.

λ	DIOR	DOTA	FAIR1M
0.1	74.2	71.3	65.1
0.2	76.8	73.1	67.2
0.3	78.4	74.4	68.5
0.4	79.1	75.1	69.2
0.5	**79.9**	**75.7**	**69.8**
0.6	79.3	75.2	69.4
0.7	78.1	74.6	68.6
0.8	76.5	73.1	67.3
0.9	74.8	71.8	65.9

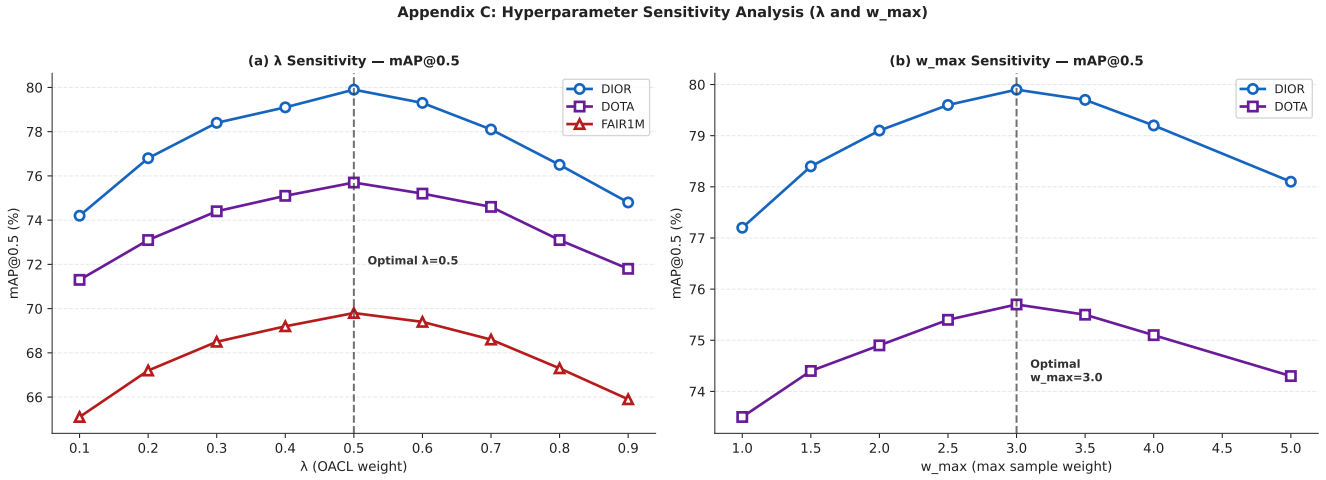


Figure 18. Appendix C: Hyperparameter sensitivity. (a) mAP@0.5 vs. λ (OACL classification weight): performance peaks at $\lambda = 0.5$ and remains within $\pm 2\%$ for $\lambda \in [0.3, 0.7]$. (b) mAP@0.5 vs. $w_{\max}$ (maximum sample weight bound): optimal at $w_{\max} = 3.0$; performance is stable for $w_{\max} \in [2.0, 4.0]$.

3. **Tugboat/Warship confusion in FAIR1M**: Fine-grained ship subcategories (Tugboat, Warship, Engineering Ship) share similar hull structures, causing persistent confusion even under optimization-aware training (Figure 11b: residual

Table 22. Sensitivity of mAP@0.5 (%) to $w_{\max}$ (maximum sample weight) on DIOR and DOTA.

$w_{\max}$	DIOR	DOTA
1.0	77.2	73.5
1.5	78.4	74.4
2.0	79.1	74.9
2.5	79.6	75.4
3.0	**79.9**	**75.7**
3.5	79.7	75.5
4.0	79.2	75.1
5.0	78.1	74.3

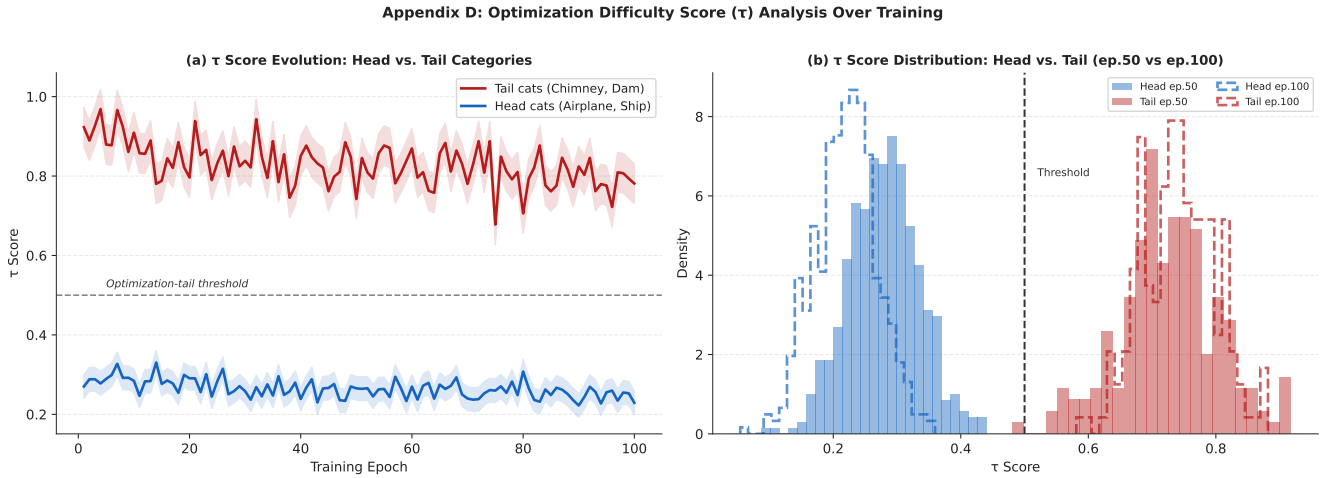


Figure 19. Appendix D: Optimization difficulty score τ analysis over training. (a) Evolution of mean τ scores for tail categories (Chimney, Dam) vs. head categories (Airplane, Ship) over 100 epochs. Tail categories maintain consistently higher τ, confirming sustained optimization difficulty. Shaded regions show $\pm$std across 3 seeds. (b) τ score distributions at epoch 50 (solid fill) and epoch 100 (step outline) for head (blue) and tail (red) categories. The dashed line indicates the adaptive threshold. The bimodal separation increases over training, confirming that τ provides increasingly reliable discrimination between easy and hard samples.

0.03–0.05 off-diagonal).

These failure modes suggest future directions: (1) incorporating global scene context (e.g., water/land classification) to resolve Harbor–Bridge ambiguity; (2) super-resolution preprocessing for sub-pixel tail objects; and (3) hierarchical classification heads that exploit category hierarchy for fine-grained ship disambiguation.